An Adequate Discovery Procedure*

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As a general rule, hypothesis is a weak kind of argument. It often inclines our judgment so slightly toward its conclusion that we cannot say that we believe the latter to be true; we only surmise that it may be so. ... The hypothesis should be distinctly put as a question, before making the observations which are to test its truth. In other words, we must try to see what the result of predictions from the hypothesis will be. C.S. Peirce (1960, 375, 380)

You see red, not infrared. Lila Gleitman

Abstract

We propose to restore the discovery procedure to the center of linguistic inquiry. To do so, however, we must abandon the goal of constructing an optimal grammar for linguistic data but pursue a merely adequate one in a sense to be precise. We propose an abductive learning procedure that incrementally generates and evaluates grammatical hypotheses and apply it to several morpho-phonological learning problems across languages in comparison to child language development and alternative, optimization-based, learning models including neural networks. This new and renewed perspective on the discovery procedure has broad implications for the pursuit of descriptive and explanatory adequacy in the study of language and its connection to cognition.

1 Discovery Procedure and Language Acquisition

To use an infinite range of linguistic expressions, children must identify the regularities residing in a finite sample to generalize beyond. In many ways, linguists face the same challenge: How do we

*†: equal contribution. This paper started during the height of the pandemic, in a large Zoom seminar on distributional learning, with vital interventions from Lila Gleitman, Norbert Hornstein, Mitch Marcus and Noam Chomsky. A preliminary model was constructed in 2021 albeit one that does not learn incrementally and cannot form structural generalizations. Portions of this work have been presented to many audiences over the years: their suggestions and feedback are gratefully acknowledged.

construct a theory of human language from only a finite body of scientific evidence? The child's and the linguist's shared task has long been noted (e.g., Sully 1895) but it is within the generative study of language where their alliance is fully ratified:

> The strongest requirement that could be placed on the relation between a theory of linguistic structure and a particular grammar is that the theory must provide a practical and mechanical method for actually constructing the grammar, given a corpus of utterances. Let us say that such a theory provides us with a *discovery procedure* for grammars (Chomsky 1957, p50-51).

As any child can learn any language(s) in the world, a discovery procedure is a theory of how children exploit their capacity for language — a general theory of linguistic structure — to acquire the specific grammars used in their environment.

It seems reasonable to assume that linguistics, like all sciences, should seek the best account of the data available. The earliest study of generative grammar states that "... the grammar must be designed in such a way as to be the most efficient, economical, and elegant device generating these [grammatical] sentences" (Chomsky 1951, p3). Similar sentiments can be found in many disciplines during that era (e.g., Shannon 1949, Kolmogorov 1965) and have continued to exert significant influence. For example, the central tenet of machine learning and AI is the rational approach, which is to develop an agent "that acts so as to achieve the best outcome or, when there is uncertainty, the best expected outcome" (Russell and Norvig 2004, p4).

Chomsky (1957) famously regarded the discovery procedure as a premature undertaking. Indeed, virtually nothing was known at the time about how children learn languages, the theory and technology of computation were in their infancy, and even the landmark Brown Corpus (Kučera and Francis 1967) was still a decade away. Two weaker conditions were proposed as alternatives. A *decision procedure* is not concerned with how a grammar is constructed but only aims to provide "a practical and mechanical method for determining whether or not a grammar proposed for a given corpus is, in fact, the best grammar of the language from which this corpus is drawn" (ibid, p51). Still weaker is the *evaluation procedure*: "... given a corpus and given two proposed grammars G_1 and G_2, the theory must tell us which is the better grammar of the language from which the corpus is drawn" (ibid, p51). Subsequent generative research developed accordingly: specify a space of possible grammars, evaluate them against the input data, and make a selection within (Chomsky 1981, Prince and Smolensky 2004).

But several decades later, we believe that a discovery procedure can now be fruitfully pursued. Large language models (LLMs), which are infamously hard to train and harder to interpret, nevertheless display impressive linguistic capacities. It is thus even more critical to develop a "practical and mechanical" discovery procedure that can be transparently related to human linguistic knowledge and its acquisition. To do so, however, we must revise a fundamental principle underlying the

study of language and learning. A theory of language need not, and in fact should not, provide an optimal account of the data: A merely *adequate* one will do, in a sense to be made precise.

As we discuss below, this alternative course of action is ultimately a reaffirmation of language as a psychological object. A discovery procedure for language is a computational mechanism confined within the limit of human cognition. From the perspective of child language acquisition, the pursuit of the best grammar not only places implausible computational demands on the learner, a point readily acknowledged in machine learning and AI (Russell and Norvig 2004, p1049), but may also display behavior at odds with how children actually acquire languages. At the same time, our advocacy for an adequate grammar reflects the long-recognized nature of language familiar to all linguists, as expressed in Sapir's adage "all grammars leak" (1921, p28). There is a limit to the orderliness of language: both the child and the linguist need to stop somewhere along the edge.

In this paper, we take some initial steps toward an adequate discovery procedure. The centerpiece of our proposal is the Tolerance Principle (henceforth TP; Yang 2005, 2016). It specifies a precise condition for linguistic productivity, the source of generalization and infinity in language. In particular, a grammar only needs to hold for a sufficiently high proportion of the data: the residual — the Sapirian leakage — can be relegated to rote memorization and no additional learning is necessary. In other words, the TP specifies a precise level of *adequacy*. As we discuss in detail, children's incremental, and in fact fairly slow, accumulation of linguistic experience provides favorable conditions for the formation and evaluation of generalizations.

The TP is often used as a decision or evaluation procedure by the linguist to validate linguistic hypotheses, often those found in the descriptive and theoretical literature, without being concerned with where these hypotheses come from. It is also possible to use the TP to iteratively form and evaluate hypotheses as shown in the application to the acquisition of dative constructions (Yang 2016, 6.2). We build on that work and present an *abductive learning* model in the sense of Charles Sanders Peirce (1934): the TP is invoked to both construct and evaluate rules incrementally in a manner similar to children's gradual accumulation of experience. The computational model is dubbed *Abductive Discovery of Productivity*, or ADP for short.[1] As discussed later, the TP is likely a generalization mechanism applicable to other domains. Consequently, the ADP model draws insights from, and maybe indeed informs, the study of categorization, learning and other cognitive processes.

Our paper is organized as follows. Section 2 reviews findings from language acquisition that are rarely considered together but collectively motivate the search for an adequate, rather than optimal, grammar. Section 3 introduces the TP and specifies the algorithmic design of the ADP model. Its properties will be illustrated with non-linguistic examples of learning and categorization to highlight the domain generality of our proposal. Section 4 presents empirical case studies of acquisition that

[1] The code and data used in the present study will be released to the public domain upon publication.

span across English, German, Icelandic, Mandarin, and Northern East Cree. Comparisons will be made against models in the optimization tradition, including neural network models, optimizers *par excellence*. We show that the ADP model provides an accurate account of cross-linguistic development, with linguistically interpretable grammars as output. Section 5 explores the implications of an adequate discovery procedure for the theoretical study of language.

2 Developmental Constraints on Discovery Procedure

A discovery procedure must provide a psychological account ("practical and mechanical method") of how children extract distributional regularities from the linguistic data. An examination of the input and the grammar that children form provide useful clues on the design specifications. Our review, as well as the empirical case studies presented in Section 4, will primarily focus on morphophonology. We do so primarily due to the easy accessibility of corpus data and the abundance of quantitative studies of development across languages. However, the lessons drawn from these studies hold for other arenas of language and its acquisition, which we discuss briefly in the concluding section of the paper.

2.1 Generalization and Non-Generalization in Child Language

The most direct assessment of children's grammar comes from the celebrated Wug test (Berko 1958). When given novel words, children can provide appropriate forms for the context (e.g., *wug-wugs* for noun plural, *rick-ricked* for verb past tense). Nevertheless, the acquisition of grammar is not instantaneous and often takes circuitous routes. English-learning children's past tense development shows a well-attested pattern of regression. In the initial stage, verbs are not consistently marked when past tense is acquired: the predominant pattern is omission (e.g., "I go yesterday"). Even in highly facilitatory conditions with familiar verbs, two-year-olds mark past tense less than 50% of the time (Schütze and Wexler 2000). The omission stage is followed by the first appearance of over-regularization (e.g., *feeled*, *goed*), indicating the emergence of productivity and generalization: the median age is 2;6 but with a considerable range of individual variation (1;8-2;11; Marcus et al. 1992). These errors are not from the input and must result from children's spontaneous creation (Ervin and Miller 1963). Concurrently, past tense marking becomes much more consistent (Kuczaj 1977). For example, "Adam," whose extensive longitudinal data has featured prominently in the study of past tense (Pinker 1995), recorded the first token of over-regularization (*feeled*) at 2;11. Prior to that point, his regular verbs are marked in obligatory context only 8% of the time but rose to 77% immediately afterwards (Marcus et al. 1992, Table A1). "Eve," well-known for her linguistic precocity, discovered the productive rule at 1;10 as evidenced by over-regularization. Prior to that point, her regular verbs are marked in obligatory context only 16% of the time but rose to 88% after (Marcus et al. 1992, Table A2). Over-regularization errors are gradually reduced over time because irregular verbs, which are not predictable and thus require rote learning, become more

firmly established in the child's lexicon. Likewise, the consistency of tense marking also improves gradually after the initial spike although children may not reach adult-level consistency until age 5 (Rice et al. 1998).

Developmental regression in morphology is very well documented across languages (MacWhinney 1978, Berman 1982, Clahsen and Rothweiler 1993, Orsolini et al. 1998, Ragnarsdóttir et al. 1999, Clahsen et al. 2002, Maslen et al. 2004). Its replication was considered a major achievement for early connectionist models (Rumelhart and McClelland 1986).[2] Less widely discussed is the nature of the regression errors. In principle, the errors may be over-regularizations, where the "-ed" suffix is extended, but also over-*irregularizations*, where some irregular form is extended, resulting in analogic errors such as *wipe-wope* and *bite-bote* along the lines of *write-wrote* and *dive-dove*.[3] Numerous longitudinal studies have established the reality of over-regularization: 4-10% of irregular verb use is regularized according to various reports, with some children reaching 25% (Maratsos 2000, Yang 2002, Maslen et al. 2004). By contrast, children almost never over-irregularize. The most comprehensive study (Xu and Pinker 1995) dubs these "weird past tense errors" on the account of their rarity. In an examination of 20,000 past-tense tokens produced by nine children, only 40 (0.2%) weird errors were found. Such a categorical distinction is also observed in experimental studies. Berko's Wug test (1958) is best remembered for testing children's knowledge of unquestionably productive processes such as *-ing*, *-ed*, and *-s*. But that study also probed whether children analogized on the basis of similarity when presented with novel words such as *bing* and *gling* that closely resemble existing irregulars. The force of analogy is exceedingly weak: only one of the 86 children supplied *bang* for the past tense of *bing* and another produced *glang* for *gling*, despite the existence of highly similar irregulars such as *ring-rang*, *swing-swung*, etc. Regular forms (*binged* and *glinged*) were the overwhelming response. Comparable results were obtained by later studies: virtually no child extends irregular forms and only regular suffixes are applied to novel items (Graves and Koziol 1971, Ivimey 1975). At the same time, Berko (1958; see also Bybee and Moder 1983) found that adults, quite unlike children, often produce irregularized forms for novel verbs ending in *ing*. Adults may approach the Wug test in a way that does not reflect their linguistic grammar (Schütze 2005). It is also possible that adults, but not children, resort to probability when no regularity is available as has been documented in similar but non-linguistic domains (e.g., Derks and

[2] It was, however, achieved under the unrealistic assumption that children initially learn a mixture of regular and irregular verbs gradually but then instantly learn hundreds of regular verbs; see Pinker and Prince (1988), Lachter and Bever (1988) for discussion. Nevertheless, this modeling decision shows that generalization only happens under an overwhelming amount of evidence, which has been referred to as the critical mass hypothesis in the connectionist literature (e.g. Marchman and Bates 1994), and is in line with the Tolerance Principle approach to learning. Current neural network models continue to struggle with developmental regression (Section 4).

[3] There is another type of error where an irregularly inflected verb is additionally marked with "ed" as in *broked* (Kuczaj 1977). These errors are due to the learner failing to deduce the verb stem given the idiosyncratic irregular forms. We do not address these error although they can be simulated on the learning models by presenting items such as *broke* as input.

Paclisanu 1967). In fact, *bing* (the Microsoft search engine) and *bling*, which entered the English lexicon after the Wug study, both take the regular past tense despite their strong similarity to the *ing* irregulars, and English has been steadily losing, rather than gaining, irregular verbs (Lieberman et al. 2007, Ringe and Yang 2022). If language innovation is primarily driven by post-childhood users (Labov 2007), then irregular forms in adult grammar are also unproductive, incapable of extending to novel items. Such asymmetries between productive and unproductive processes holds across languages. Similar to the English findings just reviewed, children's morphological errors are almost all instance of omission or overgeneralization, with virtually no instance of misusing forms with mismatched person, number or tense; see Caprin and Guasti (2009) for Italian, Clahsen et al. (2002), Mayol (2007) for Spanish, Lustigman (2013) and Bat-El (2014) for Hebrew, Clahsen et al. (2002) and Szagun (2011) for German, Gxilishe et al. (2007) for Xhosa, Demuth (2003) for Sesotho, Deen (2005) for Swahili, Allen (1996) for Inuktitut, etc.

Indeed, regressive development, reflecting the fundamental distinction between productive and unproductive processes, is not only observed in morphological learning but also in phonological (Macken 1979, Vihman 2013, Richter 2021b, Belth 2026) and syntactic acquisition (Bowerman 1988, Pinker 1989, Yang 2016). This distinction has been described by other terms: regular vs. irregular (Marcus et al. 1992), source- vs. product-oriented schema (Bybee and Moder 1983), major vs. minor rules (Lightner 1968), etc. Terminology aside, the problem children face is how to discover which form generalizes and which does not.

2.2 Experience, Variation, and Uniformity

In the generative study of language, it has been methodologically convenient to assume that speakers form a homogeneous speech community (Chomsky 1965). However, recent years have seen (re)newed interests in individual differences across child learners (Kidd et al. 2018). There is indeed significant variation in children's experience with language, which follows from both cognitive and social factors as well as other vagaries of life. It is thus even more remarkable that children nevertheless acquire very similar grammars as revealed by quantitative analyses of language variation: "The end result is a high degree of uniformity in both the categorical and variable aspects of language production, where individual variation is reduced below the level of linguistic significance" (Labov 2012, p265).

The apparent tension between variability and uniformity is best illustrated by an examination of children's vocabulary, the main focus of recent studies of individual variation (Frank et al. 2021). Children's vocabulary size is very small and develops very slowly, due to the rampant (referential) ambiguity of word meanings and the stochastic nature of the word learning process (Clerkin et al. 2017, Gleitman and Trueswell 2020). For example, a two-year-old English learner's vocabulary tops out at 500, and a three-year-old may know just over 1,000 words, and for most children, a good deal fewer (Fenson et al. 1994). Very similar growth of vocabulary size has been observed for a

wide range of languages (Bornstein et al. 2004, Szagun 2004, Hao et al. 2008).

Does a small and highly variable vocabulary result in divergent grammar acquisition? To approach this question, Jaccard similarity is used to quantify the vocabulary difference across individual children. The Jaccard similarity between two sets is the ratio of the cardinality of the intersection over the union: a higher value, up to one, indicates greater overlap and similarity. Richter (2021a) found that pairwise Jaccard similarities between the observed production vocabularies of six American English-learning children in the Providence corpus (Demuth et al. 2006) ranges from 0.25 to 0.37. We confirm and extend this basic result to the specific domain of the English verb, as this is the subset of vocabulary is directly relevant for the past tense problem. We extracted the first 200 unique verb lemmas in order of appearance for the children. The pairwise Jaccard similarity for first 50 verbs is 0.27, 0.40 for the first 100, 0.49 for the first 150, and 0.53 for the first 200.

It is evident that children's vocabularies differ considerably from each other throughout recording. Yet, their grammars formed from these verbs are generally uniform as has been amply documented in language acquisition research (Brown 1973). All children produced consistently correct basic word order, albeit with typical developmental errors such as subject/object drop, and the distinction between lexical verbs and auxiliary verbs is clear throughout. Despite very different vocabularies of verbs, and a mixture of regular and irregular verbs, all six Providence Corpus children produced over-regularized errors (e.g., *breaked*, *eated*, *throwed*).

At the same time, the largely uniform *outcome* of grammar acquisition should not overshadow differences in the developmental schedule across children. As discussed earlier, the emergence of the English past tense rule takes place as early as before 2 ("Eve") to until around 3 (Kuczaj 1976, Maratsos 2000, Yang 2016). For the six Providence children, the age of the very first instance of over-regularization ranges from 1;10 to 2;8. While it is difficult to make precise developmental predictions about any specific child, it is well documented that vocabulary size is a dominant predictor of grammatical development (Bates and Goodman 1999, Brinchmann et al. 2019). For simple systems such as English past tense, all learners appear to converge on to the same grammar. For more complex problems such as the German noun plurals (Section 4.2), however, minor individual differences have been observed across children (Gawlitzek-Maiwald 1994, Szagun 2001) that may persist into adulthood (Köpcke 1988). The interplay between vocabulary growth and grammatical development will guide our computational investigations in Section 4.

2.3 When Rules Fail

Children's impressive command of language should not obscure the fact that language acquisition does require data, and some aspects of grammar are acquired quite late (Chomsky 1969, Gleitman and Gleitman 1979, Tyler and Nagy 1989). Moreover, a productive grammar is not an inevitability and should not be imposed as a design principle on language or learning, contrary to the assumptions of leading theoretical proposals (Halle and Marantz 1993, Prince and Smolensky 2004, Marcus et al.

1992).

However, the phenomenon of gaps, widely attested gaps across languages (Halle 1973, Baerman et al. 2010), clearly indicates that productivity can, and must, be limited. Even relatively morphologically impoverished languages such as English have gaps: the past participle of the verb *stride* (Pullum and Wilson 1977) and the negative (*n't*) suffixed auxiliaries *am* and *may* (Zwicky and Pullum 1983) are absent in most varieties of English. This is so despite the fact that the majority of irregular verbs are syncretic in past tense and past participle, and the majority of auxiliaries undergo neg-contraction by adding /n̩t/ (e.g., *isn't*). Yet, neither actually emerges as productive (Yang 2017).

Child language also gives a clear illustration of the challenges that lexical gaps pose for theories of language and learning. For example, Dąbrowska (2005) shows that while Polish has two genitive singular suffixes (*-a* and *-u*) for masculine nouns, neither is the default even though *-a* is the clear numerical majority, covering over 65% of the nouns. Native speakers must resort to rote memorization of noun-specific suffixes (Mausch 2003). The acquisition of grammatical gender also reveals the limit of productivity. While some languages have gender assignment rules which children acquire early and productively (e.g. Lew-Williams and Fernald 2007 for Spanish), many languages have lexically arbitrary gender assignments, at least for a significant portion of the lexicon. A study of Icelandic gender marking (Björnsdóttir 2021), to be extensively discussed in Section 4.3, shows that gender cannot be predicted for nouns that take a null nominative singular suffix, even though neuter is the overwhelming majority: neither children nor adults show any systematic gender assignment pattern when presented with novel items. While these case studies offer direct evidence that mere majority does not guarantee productivity, the same conclusion can be reached by inspection of the familiar case of English past tense. It is difficult to be completely confident about children's vocabulary size from naturalistic transcripts, but careful investigations have repeatedly found that children learn the productive use of "-ed" only when the number of regular verbs in their vocabulary thoroughly dominates the number of irregulars, far beyond the mere majority (Marcus et al. 1992, Maratsos 2000). In this sense, gaps are simply a state of language where no rule has reached a critical mass of support for productivity: adults and children alike can only memorize the input and do not generalize beyond.

This brief review of development across languages provides clear design specifications for a discovery procedure. Broad and largely uniform linguistic generalizations must be formed on a very modest vocabulary which varies considerably across individual learners. The resulting grammar shows a categorical distinction between generalization (e.g., over-regularization) and non-generalization (e.g., omission). Gaps reflect the absence of productivity: the speaker is experience-bound, and the grammar becomes a partial function for which some input is undefined. These properties of child language acquisition pose significant challenges to learning models designed around the principle of optimality. Optimization methods generally show a high degree of sensitiv-

ity to the training data, and smoothing techniques are introduced to prevent overfitting. However, these techniques involve free parameters, including the learning models we study in Section 4, that require tuning by the researcher and have no independent empirical motivation (Roberts and Pashler 2000). Perhaps the most serious challenge comes from gaps and other failures of productivity: here, learning must grind to a halt with no generalization beyond the input. However, under the optimization-based approach to learning, there is always a best hypothesis – in terms of simplicity, probability, coverage, or a composite of all the above. To suppress it would require additional free parameters that only apply exactly when the best hypothesis fails. But child language acquisition takes place without negative evidence (Brown and Hanlon 1970, Marcus 1993): the learner therefore does not know when the best hypothesis fails and it is not even clear how to tune free parameters in the case of gaps. As extensively discussed in Section 4, the absence, rather than the presence, of productivity poses a much more serious challenge to leading theories of language and learning but has not received sufficient attention in previous research.

3 The Abductive Discovery of Productivity

We now describe the ADP model, the proposed discovery procedure that meets the design specifications of child language acquisition. Its central component is the Tolerance Principle.

3.1 The Tolerance Principle

The Tolerance Principle (TP) builds on the widely held idea (e.g., Nida 1949, Aronoff 1976, Plunkett and Marchman 1993, Bybee 1995) that productivity results from the force of regularity overwhelming exceptions. The motivation and derivation of the TP can be found in Yang (2016, Chapter 3), and we simply state the formal result (due to Sam Gutmann).

(1) Let a rule R be defined over a set of N items. R is productive if and only if e, the number of items in the set not supporting R, does not exceed $\theta_N = \frac{N}{\ln N}$.

The TP operates over the learner's lexicon measured in types.[4] For instance, if 8 out of 10 eligible items in a learner's vocabulary are attested to follow a rule, it should apply across the board because the 2 exceptions fall within the tolerance threshold ($\theta_{10} = 4$). However, if the learner learns 5 additional items but none follows the rule, the rule will fail to generalize — for $N = 15$, the 7 exceptions exceed $\theta_{15} = 5$ — and the learner will simply memorize the 8 attested examples, thereby retreating from the previous generalization.

The dynamics of learning under the TP meet the design specifications of child language acquisition reviewed in Section 2. The discrete nature of rule formation under the TP accords well with

[4] Experience is the counterpart of training data in machine learning, which obviously needs to be acquired as well in language acquisition. It can be viewed as a form of rote learning where token frequency plays a role. We put this question aside in the present paper; see Yu and Smith 2012, Trueswell et al. 2013, Stevens et al. 2017 for proposals.

the categoricity of rules in child language (Section 2.1). The proportionality criterion overcomes the variation across individual vocabularies to form a generalization (Section 2.2): as long as the number of exceptions falls on the same side of the threshold, the resulting rule will have the same behavior of generalization. Finally, a rule under the TP only needs to be adequate: when the threshold is not met, the learner will simply memorize the input (Section 2.3) and does not generalize beyond.

Under the TP, productivity is determined by two values (i.e., N and e): a categorical prediction is then made without additional degrees of freedom. By its very nature, the TP is only concerned with the coverage of a hypothesis, not with the nature of the hypothesis itself. These factors enable the TP to be applied to learning problems across domains of language (e.g., Björnsdóttir 2021, Hsieh 2021, van Tuijl and Coopmans 2021, Pearl and Sprouse 2021, Trips and Rainsford 2022, Dresher and Lahiri 2022, Belth 2023, Henke 2022, Kodner 2023, Payne 2023, Belth 2024, Li 2024, Thoms et al. 2024, Kanampiu et al. 2025, Lu et al. 2026). These studies use corpus data to approximate the learner's vocabulary and calculate productivity predictions, which are supported by converging evidence from experiments, naturalistic production data, dialectal surveys, historical sources, etc. The predictive nature of the TP can also be rigorously tested with artificial language experiments with precise manipulations of N and e for the participant's learning experience. For example, children generalize a novel plural suffix that applies to 5 of 9 words in the learning phase (Schuler et al. 2016). This is predicted by the TP as 4 is the maximum number of exceptions for 9 items ($\theta_9 = 4.1$). By contrast, a rule that applies to 3 items and has 6 exceptions, does not generalize: children typically return the singular form when queried for the plural, much like in the situation of gaps (Section 2.3). Children's contrasting behavior of generalization vs. non-generalization is difficult to capture under optimization based approaches. For example, a Bayesian model that optimizes the cost of generalization and storage (O'Donnell 2015) consistently generalizes the most frequent rule to novel items in Schuler et al. (2016)'s experiment, failing to distinguish 5/9 vs. 3/9 conditions (Bezer 2018); morphological gaps in Russian and Spanish pose similar problems. A more recent study (Shi and Emond 2023) presents a particularly stringent test. Fourteen-month-old infants who have no experience with Russian heard 16 three-word Russian sentences while playing with toys in a lab. Some of the 16 sentences are followed by a word order alternation that switches two words (e.g., ABC $\rightarrow$ BAC). For one group of infants, 11 of the 16 sentences were attested with the alternation; for the other group, only 10 were. When tested on novel sentences, the former group but not the latter recognized the alternation in the training data. The results are again predicted by the TP: $\theta_{16} = 5.8$, thus only 11/16 warrants generalization but not 10/16, despite it being the overwhelming majority. Again, the failure to generalize is notable. A transformer neural network, for example, is unable to capture the contrasting behavior under the two conditions in Shi and Emond's 2023 experiments (Friedman et al. 2026). Learning models that strive for optimality will always

have a winning form, which needs to be suppressed by some ancillary mechanism. By contrast, the TP provides a singular theory of when to generalize as well as when not to. Furthermore, because the infants have no knowledge of Russian, Shi and Emond's task is effectively one of pattern learning. Learning under the TP concerns strictly the quantitative coverage of a hypothesis rather than the nature of the hypothesis itself, providing potential evidence that the TP is a domain general mechanism of learning and decision making (Guilbeault et al. 2026).

As noted earlier, the TP is an instantiation of a traditional idea that relates high type frequency (i.e., coverage) to productivity. However, simply equating high frequency with productivity runs into problems even for English past tense. The productive rule, informally referred to as the "-ed" suffix, comprises three phonologically distinct suffixes /d/, /t/, and /əd/: among the most frequent 500 English verbs, 214 are suffixed with /d/, 145 with /t/, 89 with /əd/, and the remaining 52 are irregulars, so no suffix passes the TP threshold as an across-the-board rule. German noun pluralization, with five suffixes, is another well-known example (Marcus et al. 1995). The recursive application of the TP enables the learner to discover rules defined over specific contexts: if no rule adequately accounts for a set of words, the set is partitioned into subdivisions and the search for productivity proceeds recursively within, yielding local productive rules with narrower and more restrictive conditions of application. We describe such learning processes in Section 4 with their implications for theoretical analysis.

3.2 Abduction as Hypothesis Formation and Evaluation

Abduction, introduced by the American philosopher Charles Sanders Peirce (1877), is a mechanism for establishing beliefs and for providing explanations for "surprising facts" (Peirce 1934, 5.189). These days, the notion of abduction is usually associated with science formation, where a theory can be viewed as a belief established from data and subsequently verified and revised as additional data becomes available. However, Peirce clearly intended abduction to be a general psychological process "of forming explanatory hypotheses ... which introduces new ideas" (Peirce 1934, 5.172) and not restricted to the domain of science: "Everybody uses the scientific method about a great many things, and only ceases to use it when he does not know how to apply it" (Peirce 1877, p12). Another popular position, especially in formal epistemology and psychology, is to regard abduction as inference to the *best* explanation (Harman 1965, Lipton 1991, Rips 1989). For Peirce, however, abduction was at most inference to a *possible* explanation (Hookway 2012, p72).

We regard language learning as a kind of belief formation: a grammar is a set of particular beliefs children form based on language specific data.[5] The input data can be viewed as a set of input items

[5]Intriguingly, Chomsky (1968, p79; page number refers to the 2006 edition) alluded to Peirce's notion of abduction to describe child language acquisition. The discussion focused on the necessity of some delimitation on the hypothesis space — see Section 5 — without specifying how abduction actually takes place: "Even today, this remains a task for the future (p81)."

and their output labels, as typically formulated in the mathematical study of learning. For example, the input may be a set of words: each is specified with their phonological and morpho-semantic properties, to be mapped to an output label that denotes an inflectional suffix, a marker of word class (e.g., gender), a syntactic transformation it undergoes, etc. Traditionally, in both linguistics and machine learning, the goal is to obtain the optimal grammar that facilitates such mappings. As has been clear, the current project takes a different (and weaker) position: An adequate grammar will do, under a conception of abduction closer to Peirce's original formulation.

The main difficulty with abduction since Peirce's introduction has been the absence of a proposal about how hypotheses are formed. The epigraph from Peirce clearly indicates that not any hypothesis will do but only plausible ones ("we only surmise that it may be so ..."). In a well-known commentary, Frankfurt (1958) suggests that "What he [Peirce] must be understood to mean is that abduction leads us to adopt hypotheses *as working hypotheses*, as worthy of investigation and verification" (p595; emphasis original). But what constitutes a worthy working hypothesis is never made explicit. Peirce frequently, but unhelpfully, referred to "insight": for example, "[t]he abductive suggestion comes to us like a flash. It is an act of insight, although of extremely fallible insight" (Peirce 1934, 5.181). An exhaustive search for the hypothesis space, clearly computationally intractable, is also out of question. We propose that the TP provides the criterion of "worthy" hypotheses, which governs not only the verification of hypotheses but also their formation.

To understand the process of rule formation and verification in the ADP model (Section 3.3), consider the English verbs that undergo vowel change to /ʌ/ in past tense:

(2) cling-clung, fling-flung, hang-hung, sling-slung, sting-stung, string-strung, swing-swung, wring-wrung

There is a very noticeable pattern ("surprising fact"): most items in (2) have the vowel /ɪ/ in the stem. Here, the TP is invoked to establish the worthiness of a hypothesis: Specifically, does /ɪŋ/ constitute a *necessary* condition for the vowel change to /ʌ/? The answer is clearly Yes: for $N = 8$, 3 exceptions are tolerated, and there is only one (*hang*) in the data. Consequently, if a speaker is given a novel past tense form such as *glung* or *frung*, they would infer the stem to be *gling* and *fring* as in morphological back-formation.

Having proposed a worthy hypothesis, abduction proceeds to verification: In the epigram, "we must try to see what the result of predictions from the hypothesis will be." To do so, we need to collect *all* verbs with stems ending in /ɪŋ/ to see if, again, a TP-majority indeed changes the vowel to /ʌ/. These verbs are listed below: all are frequent enough to be known to most if not all English speakers.[6]

[6]The incremental acquisition of these items may yield minor variation along the learning trajectories (see Yang 2016, 4.1, Dressler et al. 2019 and Payne 2025 for discussion), an important point to which we return later.

(3) a. "ought": bring (1)

b. ɪ → æ/ __ ŋ: sing, ring, spring (3)

c. ɪ → ʌ / __ ŋ: swing, string, sting, fling, cling, sling, wring (7)

d. Regular: wing, zing, ding (3)

(3) contains 14 verbs. For any process defined over them to be productive, it must cover at least 9 as $\theta_{14} = 5$. However, even the most frequent process ɪ → ʌ / __ ŋ accounts for only 7 verbs, thereby failing to reach the productivity threshold. Consequently, a novel verb ending in /ɪŋ/ can only fall under the productive *-ed* suffix, and the regularity of recent verbs such as *bing* and *gling*, noted in Section 2.1, is predicted.

With this illustration on how the TP guides the formation and evaluation of hypothesis, we give the algorithmic design of the ADP model.

3.3 Recursive Abduction

ADP is an incremental learning model that receives input items one at a time and adaptively forms and revises hypothesis before moving on to the next item. For ease of exposition, we first present the algorithm in "batch" mode, i.e., a set of input items is available simultaneously, before turning to its incremental implementation.

Consider the data given in Fig. 1a. The 12 items are specified by two features — line (solid/dashed) and shape (round/square) — and are mapped to an output label of color. The patterns seem rather arbitrary and complex, but as we show below, the ADP model can be applied recursively and mechanically to uncover the regularity within while relegating the exceptions to memorization. We use this non-linguistic problem of categorization to illustrate the generality of the ADP model across domains, which has important consequences for understanding both language acquisition and the nature of linguistic structures.

The simplest hypothesis, of course, is that a single rule adequately covers all the data with a tolerable number of exceptions (if necessary). That clearly fails for Fig. 1a. The most dominant color, blue, covers 5 items, far from becoming an adequate catch-all rule: 8 would be required, as $\theta_{12} = 4$. Subdivision and recursive search are therefore required.

We propose that search pursues the largest label by cardinality. Doing so is an efficient way to achieve adequate coverage of the data and, more importantly, this strategy is supported by behavioral evidence for the animal's detection of quantity, including children's preference for dominant frequency in language learning (e.g., Hudson Kam and Newport 2009). In the current example, the model first attempts to form a plausible explanation for the blueness of the 5 objects. As in (2) and (3) earlier, the measure of plausibility is given by the TP: a property must hold for a TP-majority of items to be worthy of consideration. In other words, the TP determines the validity of a *necessary* condition for the output label. As shown in Fig. 1b, of the 5 blue objects, 4 are solid and 3 are

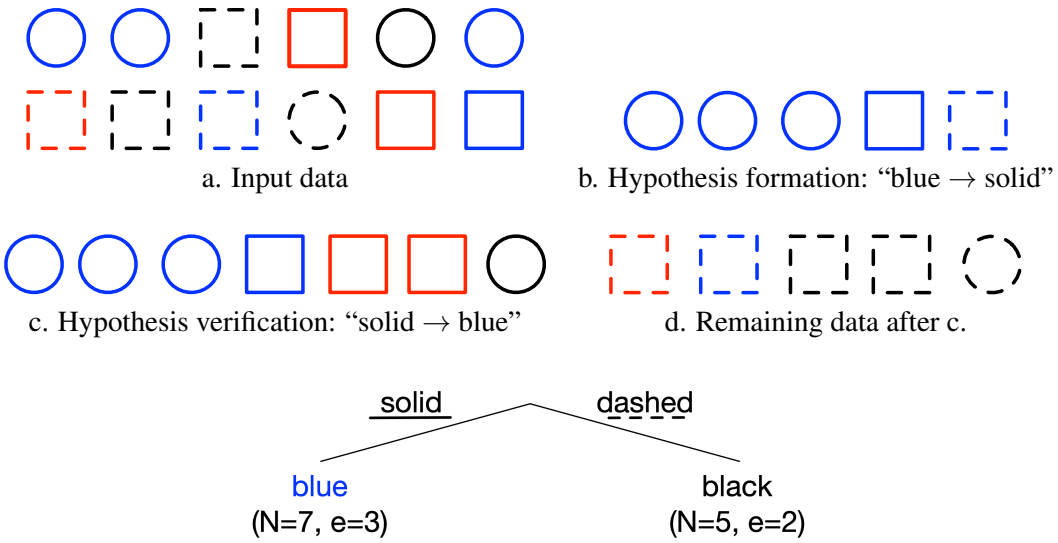


Figure 1: Recursive search for rules as a decision tree. The TP is used for both rule formation (b) and verification (c).

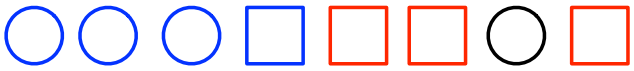
round. Here both features exceed the TP-majority threshold, but once again the model prioritizes numerical coverage and establishes “blue → solid”.

Now the model proceeds to verify whether the necessary condition, a worthy working hypothesis, is also sufficient: Namely, if an object is solid, is it predictably blue? To do so, the model must examine *all* solid items to see if enough are in fact blue by the TP. This evaluation is shown in Fig. 1c. Of the 7 solid items, only 3 are not blue: the rule “solid → blue” thus passes the TP threshold ($\theta_7 = 3$), with one black and two red items memorized as exceptions. Should the rule fail to be sufficient, the model backtracks to pursue lower ranked but still necessary conditions (i.e., “blue → circle”).

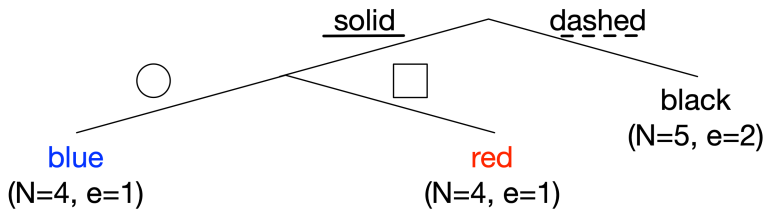
Once a productive rule is identified, the items that fall under its description are set aside, regardless of whether they follow the rule or are exceptions to it. If no productive rule for a label is found, the model proceeds to analyze the next label sorted by cardinality of coverage. The exceptions that cause an earlier search to fail may be productively accounted for by a later search. After they are set aside, the model can revisit the previously failed label, and the search of productivity may now succeed with removal of exceptions. In the present case, after learning the productive rule “solid → blue”, only 5 items with dashed borders remain (Fig. 1d). Since 3 items are black thereby passing the TP threshold, learning terminates. The two productive rules (with exceptions), are represented in a decision tree (Fig. 1e).

As discussed earlier, rules learned under the TP are provisional as they strictly pertain to the

input data that the child has internalized. The acquisition process is clearly incremental, as can be seen in the piecemeal growth of the vocabulary: ADP is similarly an incremental model, where constructed hypotheses are to be evaluated and potentially revised with additional input. When a new item is learned, the model finds the appropriate branch of the decision tree based on its features and adjusts the TP measures for the corresponding rule. For example, if a new item [solid, square] is blue as predicted, then the left branch rule in Fig. 1e will be updated as ($N = 8, e = 3$): the coverage gets better while the number of exceptions remains the same. But a rule can also be overturned and revised if additional input tips it below the TP threshold.



a. The final item [solid, square] is red, breaking the rule in Fig. 1e.



b. A Revised decision tree.

Figure 2: Incremental learning of items and refinement of rules: (a) breaks a rule and triggers a revision (b).

Suppose the next item, [solid, square], is in fact red as shown in Fig. 2a. The rule "solid→blue" is broken: for N=8, only $\theta_8 = 3$ exceptions are allowed. The ADP model is triggered recursively, as it seeks to identify subdivisions within the set of solid items. Now the shape feature is implicated. It is easy to see that the same procedure described above will produce two sub-rules defined over "square" and "circle" respectively, both of which are adequate (Fig. 2b).

Fig. 3 provides a high-level description of the ADP algorithm. ADP yields a decision tree: the leaves of the tree are rules defined by the sequence of features. In the incremental learning model, a new input item will be located to a rule by traversing a sequence of features — say F_1, F_2 ... F_i — increasing the specificity of rule conditions. If the matching rule ceases to be productive by the TP, the model traverses up to the immediately preceding node corresponding to feature F_{i-1}, and attempts to identify a productive rule by calling the ADP algorithm again (Fig. 3). If no such feature can be found, the model traverses upward yet again to the node with feature F_{i-2}, and so on. In the limiting case, a single exception may break all decision branches, and learning in effect starts over. The branches correspond to linguistic features: closer to root are those established fairly early, with refinement branches added when the data is further partitioned.

The difference between batch and incremental learning deserves emphasis. Batch learning is

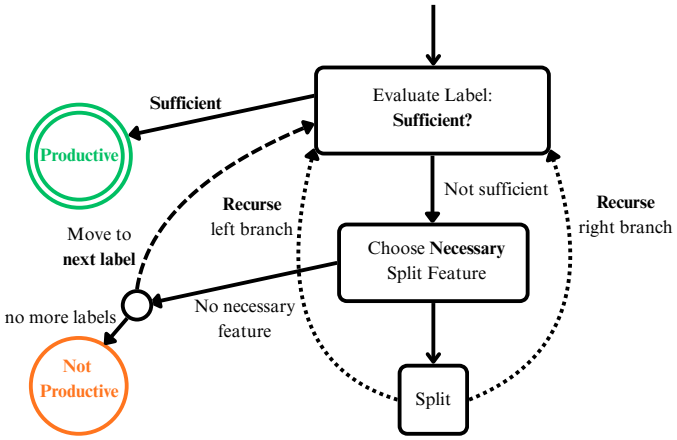


Figure 3: A schematic illustration of the ADP algorithm and the recursive search of productive rules.

akin to how an analyst approaches a linguistic problem: the data is gathered and an analysis is postulated for its totality. Incremental learning reflects child language acquisition, where the vocabulary grows gradually and hypotheses are provisional. The two may arrive at different grammars from the same data. Batch learning of the 12 items in Fig. 1a yields the two rules in Fig. 1e, but the same items presented incrementally in the order shown yield three productive rules. As the reader can verify, the model can learn "circle→blue" ($N = 5, e = 2$) and "square→red" ($N = 6, e = 3$) after 11 objects, but the last blue object ([solid, square]) breaks the latter rule, triggering a split into "[square, solid]→red" ($N = 3, e = 1$) and "[square, dashed]→black" ($N = 4, e = 2$). Since the order of presentation corresponds to an individual learner's vocabulary growth, ADP can achieve a high degree of uniformity in the learned grammars (Section 2.2) while leaving room for variability across individuals. And there may be incongruence between the grammar children discover incrementally and the grammar linguists postulate in batch: English past tense allomorphy (Section 4.1.4) will provide a case in point.

3.4 Efficiency of Learning by Need

ADP is best viewed as a metaheuristic algorithm: it provides a mechanism by which hypotheses are constructed and evaluated but is independent of the nature of the hypotheses, because the TP is concerned with the quantitative coverage of the data (N and e). In principle, alternative formulations of a learning threshold can be incorporated into the ADP architecture as well. The learning process illustrated by the label coloring example therefore applies equally to language acquisition, with a different set of features and hence a different hypothesis space. Its application to language nevertheless takes a stance on some important issues in the theory of language, cognition, and learning; we focus here on how the model forms generalizations from items and return to the tension between form and substance in the concluding section.

The ADP model can be viewed as an algorithmic synthesis of psychological and linguistic studies of learning in the past few decades, including the TP. As discussed earlier, the pursuit of larger labels is motivated by frequency effects in language acquisition but also by the fact that rules covering more items are more likely to appear earlier in the input. The one-feature-at-a-time strategy in hypothesis formation has the clear benefit of restricting the hypothesis space. It is additionally motivated by findings that both adults and children attend to a single attribute of information in learning and categorization (Nelson 1984, Medin et al. 1987, Smith 1979) but do not seek to maximize overall similarity that involves multiple features. Therefore, ADP searches through the list of features one at a time, again favoring coverage, to propose potential hypotheses. Consequently, each of the multiple features implicated in a rule must be learned sequentially, with the later ones refinements (i.e. splits) of earlier ones. Search stops as soon as a feature is found to yield an adequate rule. The total number of hypotheses formed and evaluated for each incoming input item is therefore bounded by the number of total input items multiplied by the cardinality of the feature set, keeping computational effort at a minimum.

The rules in the shape-coloring example are defined over atomic features (circle vs. square, solid vs. dashed). In language, rules may be defined over atomic features, but may also make use of abstractions. Take, again, the case of English past tense "-ed". Following a well-known discussion by Halle (1978), the voiceless consonant /x/ as in the German word *Bach* does not appear in English, but English speakers would use /t/ in past tense formation ("He out-Bached Bach"). This is only explained if the choice of the suffix is conditioned on the voicing feature of word-final segments: not a list of segments but a generalization over segments. At the same time, not all phonological rules are based on abstractions: phonological systems can also be irregular at the level of segments and individual words, as in the tensing of /æ/ in Philadelphia English before a set of consonants that does not form a natural class (Labov 1989, Labov et al. 2006), a system that children nevertheless acquire in due time (Roberts and Labov 1995).

The construction of feature-based rules in ADP once again illustrates the principle of adequate learning. The consideration of abstractions applies when ADP's input contains segmental phonological forms (e.g., the lemma /wɔk/ *walk* in the English past tense), though the same procedure could be extended to other kinds of representations (tone representations, etc.). ADP attends to the segment of a phonological form adjacent to what is being predicted (e.g., the stem-final segment when predicting a suffix or the initial segment when predicting a prefix). It seeks a set of segments that are individually productive (sufficient) predictors of the label currently being pursued and that together cover a necessary number of the items with that label. If the individual segments provide such a predictive set, ADP can use them as a new feature for constructing a new hypothesis. Generalization over combinations of distinctive features (natural classes) is only attempted if the set comprises more than one segment. If the adjacent segments fail to productively predict the label, ADP constructs a

new *tier*-like representation using Belth (2024)'s mechanism: any representation-adjacent segment that is not a productive predictor is excluded from the tier. The search for productivity continues recursively on the new representation; the recursive removal of segments from the representation incrementally brings long-distance dependencies into view locally. This change of representation occurs only if an adequate generalization cannot be found over the current representation.

This representation-change operation is related to formal theories that attempt to delimit the expressive power of phonological processes (Kaplan and Kay 1994, Heinz 2018), in particular tier-strictly-local processes (Burness et al. 2021). In our view, because both the creation of new representations and generalization over them invokes adjacency, the operation may also reflect the more general principle of perceptual learning and pattern recognition that favors local dependencies over non-local dependencies, as has been observed in both infants and adults across domains (Gómez 2002, Gebhart et al. 2009) and non-human species (Wilson et al. 2020). Importantly, new and/or more abstract representations are only pursued when adjacent information does not result in productive rules. Again, a rule need not be perfect: If a TP-majority of words can be accounted for by a rule defined over items adjacent on the current representation, then those that cannot can be memorized without triggering new representation construction (Belth 2023, 2024).

Taken together, learning under ADP is driven by the need to establish productive rules. Refinement of a rule — by subdivision, by backtracking, by forming abstract representations — only takes place when the rule fails, which keeps computational cost low and offers ways to incorporate cognitive constraints into the learning process. The division of labor between formal and substantive universals within a theory of language (UG), and between a domain specific UG and domain general mechanisms of learning, is taken up in Section 5. For now, let us turn to some major empirical problems in child language acquisition.

4 Modeling Rule Learning Across Languages

We now present five empirical case studies of language acquisition: English past tense inflection (Section 4.1), German noun plural suffix allomorphy (Section 4.2), Icelandic gender assignment (Section 4.3), Mandarin noun classification (Section 4.4), and Northern East Cree possessive suffixation (Section 4.5). They are also accompanied by extensive empirical child language research that has had major impact on linguistics and cognitive science. These problems represent a broad typological range, and the grammatical rules involve choices over phonological, morphological, and semantic properties that vary across languages. We show that the ADP model discovers these rules across the typological range, producing linguistically interpretable rules while also capturing the major developmental findings in child language.

As a comparison, we evaluate two classes of alternative learning models which, despite their very different theoretical orientations, both embody the optimizing approach to language and learn-

ing. The Minimal Generalization Learner (MGL; Albright and Hayes 2003) forms rules of varying granularity over input-output pairs. The rules are evaluated according to the scope and accuracy of coverage, and further adjusted to obtain a confidence score, modulated by two hyper-parameters. For a novel item, the rule with highest confidence score is applied to generate an output. We also tested three neural network models, which differ in architecture but are modern optimizing learners par excellence. The KC model (Kirov and Cotterell 2018) is an LSTM encoder-decoder network specifically proposed as an update of the Rumelhart-McClelland model (1986) that launched the connectionist approach to language acquisition. CHR-TRM (Wu et al. 2021) is a character-level transformer used as a baseline for recent NLP tasks (e.g., Kodner et al. 2022), and CLUZH (Wehrli et al. 2022) is a character-level transducer specifically designed for low-resource settings similar to child language acquisition. Like ADP, the neural models can operate over arbitrary features. The MGL model is designed to operate over phonological segments and features although it can be augmented in a very limited way (Albright 2002) to be discussed later. Finally, unlike the ADP, both MGL and neural models have hyper-parameters that require tuning, which will also be discussed later. The comparison models are evaluated on English, German, and Icelandic; for Mandarin and Northern East Cree, whose datasets are much smaller, only ADP is used (Section 4.4).

Both MGL and the neural networks, like the ADP (and the TP), operate over word types in the lexicon rather than word tokens (see footnote 4), a departure from statistical learning models in NLP and from early connectionist models. The change reflects the consensus reviewed earlier that type frequency determines productivity and generalization, and it remedies a well-known problem: trained on tokens, the networks are dominated by the highly frequent irregular verbs and produce many bizarre errors (Pinker and Prince 1988), including massive over-irregularization that almost never appears in child language (Marcus 1995, Plunkett and Juola 1999); see Nikolaev et al. (2026) for a head-to-head comparison with token-based training (Baayen 2009). However, as we will see, these problems persist. An optimizer, whether defined on rules that generate word forms (such as MGL) or on word forms directly (such as the neural models), will always have a highest-valued output; for human learners, the correct option in many cases is to have no output at all (Sections 2.1 and 2.3).

4.1 English Past Tense

Our first empirical study models the acquisition of English past tense, one of the most extensively studied topics in linguistics and cognitive science.

4.1.1 Models and Experimental Setup

In a typical study of computational learning, models are constructed on training data, then calibrated on validation data to combat overfitting and enhance robustness, before being evaluated on held-out test data. We follow such practice in this paper, but a more prominent focus will be on the

developmental process of the learning models. We will describe the operational details for the English past tense experiments; the minor variation for the other case studies will only be briefly summarized.

To model the learning of English past tense, we first extracted the verb past tense forms from the North American English section of the CHILDES database (MacWhinney 2000). To minimize the transcription errors in CHILDES annotation, these verbs were intersected with the English sections of combined UniMorph 3 (McCarthy et al. 2020) and 4 (Batsuren et al. 2022) and converted to IPA with the CMU Pronouncing Dictionary. In all, 1,800 verbs were obtained. Following best practices in machine learning, we held out a random sample of 300 words for our test set, while our training set contained 1,500 total words. The words are represented as input-output pairs. Their phonology is represented as a sequence of segments, each of which is specified by an array of phonological features as specified in a broad coverage database (Mortensen et al. 2016): For example, the verb *call*, represented as /kɔl/, and its inflectional feature (past) are mapped to form /kɔld/ (*called*).

To simulate the incremental nature of child language acquisition, we randomly sampled, without replacement, the past tense forms of the 1,500 training words with probabilities proportional to their token frequencies, thereby obtaining a sequence of verbs approximating the growth of vocabulary that favors more frequent items (Swingley and Humphrey 2018, Braginsky et al. 2019, Kodner 2019). This sampling scheme produces Jaccard similarities across the simulated vocabularies comparable to the empirical measures obtained in the Providence corpus (Section 2.2; 0.27/0.40/0.49/0.53 at 50/100/150/200 verbs). 10 such training sequences ("seeds") were independently sampled to represent 10 virtual learners. While ADP is very efficient and learns in a matter of seconds, both MGL and the neural networks are iterative optimization models and training may take days even for a few hundred words: 10 virtual learners are the practical limit of the computing resources at our disposal.

The ADP model was trained strictly incrementally, processing one word at a time in the order the words were sampled. Neither MGL nor the neural networks are incremental learning models. To simulate the effect of incremental training and to assess their suitability as models of child language acquisition, these models were trained in batch for each seed with checkpoints of every 100 words: that is, the models were trained on the first 100 sampled words, then with the next 100 sampled words in addition, and so on, such that each next largest training size is a superset of the previous training size. These checkpoints then mirror the training data accessed by the ADP model as closely as possible. For each training size, we split the seed data randomly into 80% training and 20% finetuning (e.g., the first 100 training items were split in 80 train and 20 finetune, the full 1,500 training items were split into 1,200 train and 300 finetune). MGL's upper and lower confidence hyperparameters were tuned by a grid search over the values *lower confidence* $\in \{45, 50, 55, 60, 65\}$ and *upper confidence* $\in \{85, 90, 95, 99\}$ using accuracy on the finetune set to choose the best set

for each training size and seed. For CHR-TRM and KC, we used the hyperparameters suggested by the original authors for small training conditions (Wu et al. 2021, Kirov and Cotterell 2018). For CLUZH, we used beam decoding (beam size 4) and the large training hyperparameters from Kodner et al. (2023b) for all languages rather than those in the original study which employed language-specific hyperparameters. For a given train size, the 20% of finetuning data was used for hyperparameter search (MGL) and as the "dev" set for training the neural models. ADP requires no fine-tuning: the entire training data was used for learning.

A model's prediction for an input item (e.g., verb stem) is compared with the correct past tense form. For ADP, prediction is only made if the stem falls under a productive rule; otherwise the input item itself is returned, mirroring children's omission of marking before the acquisition of productive rules in natural languages (Marcus et al. 1992) as well as in experiments when presented with novel items for which no productive rule exists (Berko 1958, Schuler et al. 2016). MGL and the neural models, which do not make a categorical distinction for productivity, produce the best candidate for each input stem: for MGL, the rule with the highest confidence score is applied, and for the neural models, the output form with the highest probability. As can be expected with most modern machine learning studies, all models become increasingly accurate on the held-out test set as the amount of training data increases, and all exceed 90% accuracy (averaged over the 10 seeds) by the end of training, although they differ in the speed of learning and in the shape of their trajectories. Furthermore, the standard machine learning pipeline of train and test only evaluates the terminal state of models: accuracy must be accompanied with systematic error analysis to assess learning outcomes and diagnose deficiencies (Gorman et al. 2019, Kodner et al. 2023a). Developmental details are even more critical in a cognitive setting of language acquisition, which will be the primary focus of our study.

4.1.2 Developmental regression

As reviewed earlier, the emergence of over-regularization in English past tense is one of the most important findings in language acquisition. Probing for such errors in computational learning models, however, is less straightforward. In natural language acquisition, children often alternate between correct and incorrect forms for irregular verbs (e.g., *goed* and *went*). Because the correct form for the irregulars must be rote-learned, the failure of memory retrieval yields over-regularization when a productive rule is applicable. Computational learning models, however, typically have infallible memories. Given the vast number of connections and parameters, neural network models are typically very good at memorizing the training data (Zhang et al. 2021). For ADP, input items are either accounted for by productive rules or lexically memorized as exceptions: no failure is possible. The design of MGL, as we will see, makes learning irregular verbs difficult, albeit not in a way similar to human children. While it is interesting to study the gradual consolidation of irregular forms (Pinker 1995, Yang 2002, Regel et al. 2015), the focus of the current study is to understand the emergence

of rules and associated developmental changes such as regression. To this end, we measure the developmental trajectories in the following way. At every developmental check, we computed the percentage of correctly inflected forms for *all* verbs: those that have been used for training can be expected to be correct for most models but those that have not appeared effectively constitute a natural Wug test, which corresponds to the situation when children learn new words. Following the study of children's past tense, the no-change verbs are excluded from evaluation (Maratsos 2000, Maslen et al. 2004) because it is impossible to distinguish the correct past tense forms from unmarked forms. The results are reported in Fig. 4.

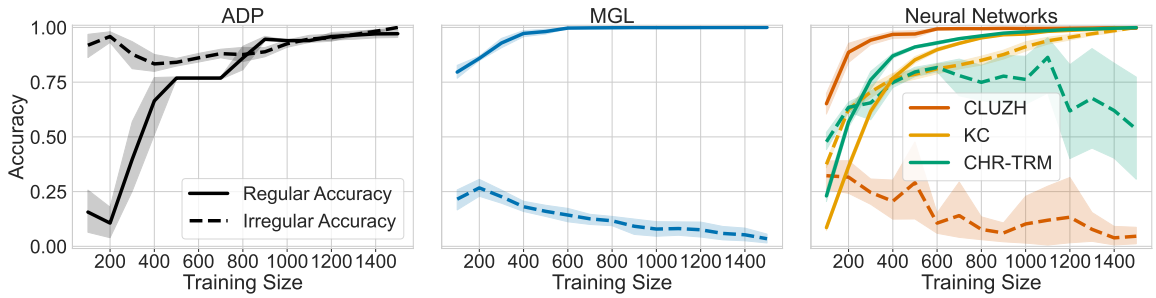


Figure 4: Developmental trajectories for inflected regular (solid line) and irregular (dashed line) verbs over 10 seeds; both training and testing verbs are included.

ADP shows a clear pattern of developmental regression: the accuracy of irregular verbs dips before gradually increasing again. We invite the reader to inspect the ADP learning trajectories across the seeds (i.e., simulated individual learners) at https://adp-li.netlify.app/ where the results from German plurals (Section 4.2) and Icelandic gender (Section 4.3) are also reported. It is notable that the dip consistently takes place around 200 verbs in the lexicon. Using 15% as a proportional measure of verbs in the MacArthur-Bates Communicative Development Inventories (Fenson et al. 1994), we can infer the median age by which English-learning children acquire 200 verbs to be about 30 months based on Mayor and Plunkett's 2011 correction for undersampling (Table A3), which closely aligns with the empirical finding of 2;6 based on children's first over-regularization errors (Marcus et al. 1992). The accuracy of regular verb marking increases dramatically just as over-regularization of irregulars arises, consistent with their concurrent development in child English (Marcus et al. 1992; Section 2.1). Both MGL and the transformer models (CLUZH and CHR-TRM) have difficulty learning the irregular verbs; accuracy in fact gets worse with more training data. As usual, it is difficult to understand precisely how neural network models work and why they fail, but the transparency of MGL explains why it struggles with irregular verbs. MGL learns lexically specific rules for irregular verbs. However, the regular rule reaches a higher confidence score even when the vocabulary size is still very modest because the number of regular verbs already thoroughly overwhelm all irregular groups: even the irregular verbs in the training data get regularized. The KC model does succeed in learning irregular verbs in the input but shows a monotonic trajectory

without evidence of regression.

4.1.3 Distribution of errors

To further understand the developmental patterns in the learning models, it is important to analyze and compare the types of errors they produce against those produced by children. As reviewed in considerable detail (Section 2.1), there are three logically possible errors: (a) omission where no past tense marking is used when marking is obligatory (e.g., "I bite it already"), (b) over-regularization where the regular form is used for an irregular verb (e.g., *bite-bited*), and (c) over-irregularization where an incorrect irregular form is used by analogy (e.g., *bite-bote*, *wipe-wope* on the lines of *write-wrote*, *drive-drove*, *ride-rode*, etc.). The early stages are dominated by omission. After the emergence of productivity, over-regularization of irregulars is robustly attested but over-irregularization is virtually absent, a general pattern that holds across languages.

Fig. 5 gives the error analysis for the computational models. Like the developmental regression study in Section 4.1.2, all training verbs are used, again with the exclusion of the no-change irregulars: those unseen by the checkpoints serve as Wug tests, providing the models opportunities to make errors.

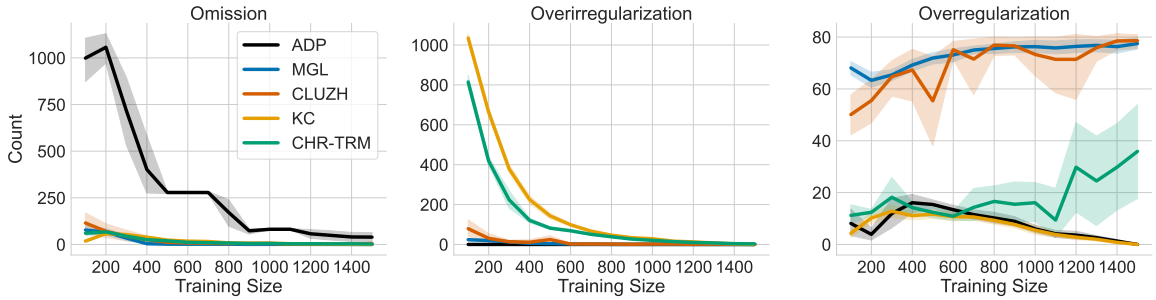


Figure 5: The distribution of error types among omission, over-irregularization, and over-regularization. We give error counts, rather than percentages, for visual convenience as only a very small portion of the verbs, namely the irregulars, are subject to over-regularization while all verbs are subject to omission and over-irregularization.

ADP is the only model that reproduces the distribution of errors in child language. First, as shown in Fig. 5a, only ADP shows an initial dominance of omission followed by a steep decline. Optimality design explains why omission errors are difficult to obtain in the other models. There is always a best output, whether from rule confidence scores for MGL or from predictive probability: no output is not an option.[7] Second, as shown in Fig. 5b, ADP, like children, virtually never over-irregularizes, while KC and CHR-TRM over-irregularize very frequently. Although MGL and CLUZH do not over-irregularize frequently (Fig. 5b), they do over-regularize albeit far too early –

[7]The no-change rule is indeed a pattern that the models learn. Its over-application as a form of irregularization would create the appearance of omission but the rule applies to just a handful of irregular verbs and can never emerge as the best option.

even at the very beginning (Fig. 5c). Again, MGL's behavior is interpretable. As discussed earlier, the regular rules receive the highest confidence scores very early, effectively as soon as the regular verbs become the majority of the lexicon. However, as has been widely documented, the initial stage of past tense learning is lexically based, and over-regularization only emerges when the number of regular verbs has reached a critical mass beyond majority.

The behavior of ADP is also easily interpretable. By definition, unproductive rules do not generalize at all. Therefore, a dominant stage of omission and the absence of over-irregularization are two sides of the same coin. Indeed, the precipitous drop in omission just after 200 verbs corresponds to the emergence of over-regularization in Fig. 4. Recall the verbs that end in "ing" in (3): even the most frequent rule (vowel change from /ɪ/ to /ʌ/) fails to reach the TP threshold. The same holds for every irregular pattern: the absence of over-irregularization is predicted.

4.1.4 The independence of allomorphy in acquisition

English regular past tense is often a textbook example of morpho-phonological alternation (e.g., Hayes 2008). The suffix is typically described as a single underlying form (/d/; Halle and Marantz 1993) that surfaces as /t/ following voiceless consonants via assimilation. The epenthetic vowel insertion in /əd/ is regarded as an automatic response to general phonotactic constraints (McCarthy 1986, Baković 2005). Similar analysis can be given for the English noun plurals (e.g. Kenstowicz and Kisseberth 1979), where the surface forms (/s/, /z/, /əz/) are derived from a single underlying form (/z/).

While accurate and elegant, this characterization of English past tense (and plural) allomorphy is at odds with how children actually acquire the system. If three suffixes are indeed alternative realizations of a single underlying form, with voicing and epenthesis reflexes of the phonological system, their acquisition should exhibit synchronized uniformity — at least on a strong interpretation of the theory, which has the effect of reducing the learning of three separate processes to one. However, it has been known since Berko's original Wug test (1958) that the acquisition of /əd/ takes much longer, and not even /t/ and /d/ suffixes are learned at the same time. For example, the children in Berko's study inflected *rick* with /t/ 73% of time, *spow* with /d/ 52% of time, but *mot* and *bod* with /əd/ only 33% and 31% of time. Elicitation studies also show a significant delay for /əd/ in comparison to /t/ and /d/ (Tomas et al. 2017).

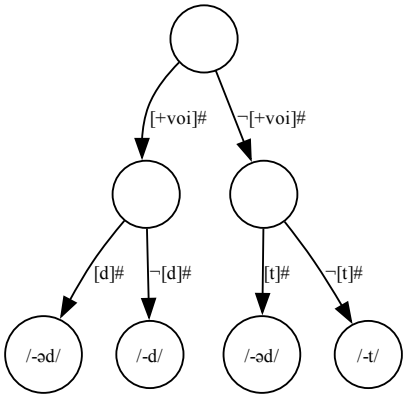
The wealth of naturalistic production in CHILDES provides direct support for the independent developments of the past tense suffixes. We examined an irregular past tense corpus of children learning North American English in CHILDES, which includes over 53,000 tokens, the largest collection to our knowledge.[8] We investigated the emergence of productivity for 25 [t/d]-final (TD) and 65 non-[t/d]-final (NTD) irregular verbs by examining the data from 100 children who (a) over-regularized and (b) have longitudinal data of at least six months or more. Overall, 96 of the

[8]We thank Christine Soh Yue for providing us with this dataset. Once again, no-change verbs are excluded.

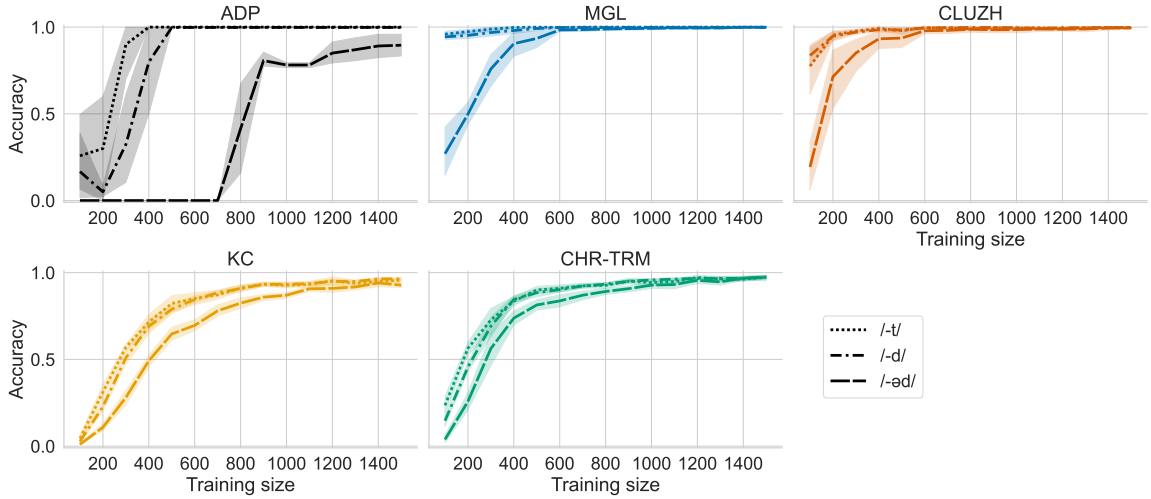
100 children over-regularized NTD verbs, with an average error rate of 8.43% (1124/13332), and only 22 over-regularized TD verbs, with an average error rate of 1.66% (142/8561). The error rates for the NTD verbs are significantly higher than TD verbs ($\chi^2(1) = 438.8, p < 0.001$). The difference is not due to the higher frequency of the TD verbs in past tense. If that were the case, the rote-learning of TD verbs would be more secure than NTD verbs, thereby providing stronger resistence to over-regularization. However, the frequencies of the two verb groups did not differ in children's speech (Mann-Whitney test $U = 724, p = 0.43$) or in caretaker speech (Mann-Whitney test $U = 866, p = 0.81$). The error rate difference is explained if /əd/ becomes productive later than /t/ and /d/: TD irregulars face a productive competitor (e.g., /əd/ in *bended* and *writed*) later than NTD irregulars (e.g., /t/ in *thinked* and /d/ in *comed*). That is, by the time TD irregulars become eligible for over-regularization, they are already better consolidated in rote memory. Furthermore, there is also individual-level evidence for the late emergence of /əd/. 18 of the 100 children over-regularized both TD and NTD verbs. The mean age of first over-regularization is 3;0 (NTD) and 3;7 (TD): the difference is statistically significant (two-sided paired t-test, $t(17) = 3.48, p = 0.003$). These analyses establish the independent development of English past tense allomorphy, which has not been systematically investigated before.

ADP provides a transparent account of how and why the three suffixes /d/, /t/, and /əd/ emerge along the incremental acquisition of the vocabulary. As noted in Section 3.1, no suffix is sufficiently dominant to tolerate the others as exceptions, which necessitates the subdivision of the verbs for recursive abduction. The voicing feature is easily identified as a necessary (and in fact perfect) criterion for separating /t/ and /d/, but it cannot reach productivity as a sufficient condition when the learner's vocabulary is small. The exceptions include not only the irregular verbs but also the regular verbs that end in /t/ and /d/ but take /əd/. Only when the vocabulary reaches a certain size are /t/ and /d/ suffixes learned: /t/ is learned slightly earlier because it is only counteracted by irregular verbs that end in voiceless consonants whereas /d/ is also affected by irregular verbs that end in vowels. But the productive condition for /əd/ is delayed further still in order to overcome irregular verbs that end in /t/ and /d/. The decision tree produced by the ADP model is shown in Fig. 6a, which transparently maps onto the structural descriptions familiar to linguists. The condition of allomorphs, such as the implication of voicing in *out-Bached*, is correctly identified when learning stabilizes.

The developmental trajectories of the three suffixes, indicated by the size of the vocabulary consumed by the learning models, are shown in Fig. 6b. ADP shows a clear differentiation: /t/ precedes /d/, and both are ahead of /əd/, consistent with the acquisition findings. This does not mean that the traditional analysis of a single underlying representation of the past tense morpheme is incorrect, but it must be justified on other grounds which would require a consolidation process that unifies three independently acquired alternatives sometime later in the learner's life. By contrast,



(a) Decision tree for English past tense learned by ADP.



(b) Development of /-t/, /-d/ and /-əd/ in the learning models.

Figure 6: The representation and development of English past tense allomorphs.

two neural models (KC and CHR-TRM) show little differentiation. MGL and CLUZH do learn /əd/ later than /t/ and /d/ but still far too early in comparison to children.

In sum, all learning models eventually reach high accuracy on unseen data, but only ADP, but not the other models under tested configurations, characterizes children's past tense development (Sections 4.1.2 and 4.1.3), and the modeling and acquisition results on allomorphy challenge the traditional structural analysis (Section 4.1.4). The child's arrival at the target grammar is never in doubt; it is the path to the target that constitutes the object of study, and computational models that lay claim to language acquisition should be evaluated on that path rather than on the terminal state of training alone (Hu et al. 2020, Warstadt and Bowman 2022, Wilcox et al. 2024; see Martínez

et al. 2023). This theme will be revisited in the case studies that follow.

4.2 German Noun Plurals

The German noun plural problem first gained prominence in language acquisition during the past tense debate (Marcus et al. 1995). Nouns in German form plurals with several suffixes: *-s*, *-n*, *-en*, *-e*, *-er*, and the null *-ϕ*. Of particular note is the *-s* suffix: it covers the smallest fraction of nouns (about 2-3%) but is nevertheless productive as can be seen in newly formed nouns (e.g., *iPhone-iPhones*). Contrary to the dual-route model (Pinker 1995, Clahsen 1999), according to which only *-s* is productive and all other suffixes are memorized by associative memory, most theoretical accounts hold that multiple suffixes are productive as conditioned on gender and/or phonology of the noun (Wiese 1996). Multiple suffixes are over-applied by young children (Elsen 2002, Köpcke 1998), the clearest evidence for their productivity. Unlike the simpler problem of English past tense where all learners converge to the same grammar, the complexity of German plurals results in a larger range of individual variation during language acquisition. Yang (2016, 4.4) presents a recursive application of the TP to the problem. The results confirm the productivity of several suffixes, consistent with empirical and theoretical descriptions (Trommer 2021), but that study cannot be fully regarded as a computational model of acquisition. For one thing, the 500 nouns used are analyzed collectively ("batch" mode), rather than in an incremental fashion as in child development. For another, the rules considered were taken from the descriptive literature and subsequently validated. That is, the TP is used as an evaluation/decision procedure rather than a discovery procedure.

To test the learning models, we extracted noun plurals from the Leo Corpus (Behrens 2006), which were then intersected with the combined Unimorph 3 and 4 morphological corpora for further validation. The extracted singular-plural pairs, 2000 in all, were transcribed automatically with the Epitran library (Mortensen et al. 2018). Following prior modeling work (McCurdy et al. 2020, Dankers et al. 2021), umlauts were excluded. The focus of the current study is on the acquisition of the suffixes, and umlaut is understood to be a separate phenomenon although its dependence on the suffix selection (Wiese 1996) can be further quantified by TP/ADP; see Belth (2026). In addition to phonemes, each noun in the data is further annotated with its grammatical gender (masculine, feminine, and neuter). It is well documented that gender marking is acquired very early on a lexically specific basis from gender-marked determiners (Szagun et al. 2007) and accuracy reaches over 90% by age 3 (Kupisch et al. 2022). Again, the data was sampled to create 10 seeds (virtual children), with each split into train, finetune, and test following the same procedure in the preparation for the English data.

By design, the MGL model only operates on phonological representations, though a single feature of gender can be added as a "hack" (Albright 2002, p.72) by inserting it as an additional "segment" in the phonological representation. The neural models, like ADP, can operate on arbitrary features as input. As in the case of English past tense, all models achieve good results on held-out

test items when the input data volume increases. This aligns with the descriptive studies where it is reported that 80-85% of German noun plural are predictable from the properties of the singular (Hahn and Nakisa 2000, Elsen 2002). However, as we have seen in English past tense, predictive accuracy may not be informative about their suitability as models of language acquisition.

Compared to English past tense, there have been many fewer longitudinal studies of German plural acquisition: the system is far more complex than English and there is comparatively little naturalistic data in the public domain. The most comprehensive study uses an elicitation task with a cross-sectional population (Laaha et al. 2006), where 84 German-speaking children (2;6 to 6;0) in 7 age groups were asked to provide the plural form for 42 pre-selected German nouns that provide a balanced coverage of all plural suffixes. The approach is arguably more revealing than naturalistic data that provides little opportunity to observe children's knowledge of the rarer suffixes. After ruling out cases where children, predominantly the youngest group, supplied a singular form when a plural form is demanded, over-generalizations (1,305 tokens, recalculated from Laaha et al. 2006, p293-4) were found for all suffixes but at very different rates. No-marking is the most frequent (43%), followed by *-e* (24%), then *-(e)n* (15%) and *-s* (14%), with *-er* overused only 2% of the time. This dataset enables us to evaluate the learning models on the 42 test nouns, which are available from the study, to see how well they match children's suffix error distributions. We tested the predictions of the learning models on the 42 nouns for all the vocabulary checkpoints and reported the average proportion of suffix over-application, which can be viewed as a simulation of young children's performance across the age span in Laaha et al. (2006)'s study.

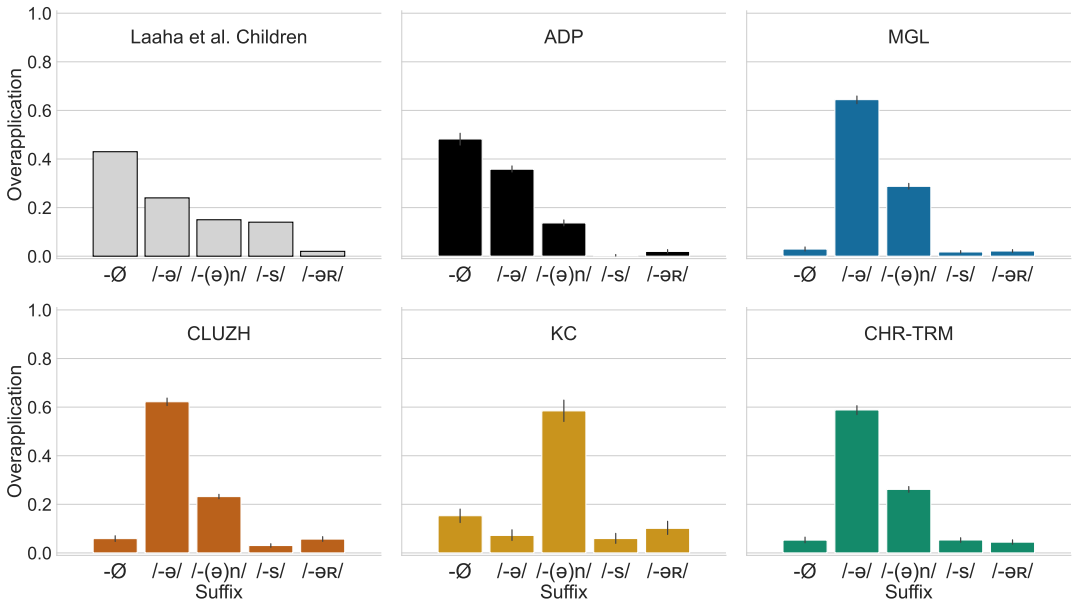


Figure 7: Overapplication of plural suffixes on 42 nouns from Laaha et al. (2006) in comparison with German-learning children.

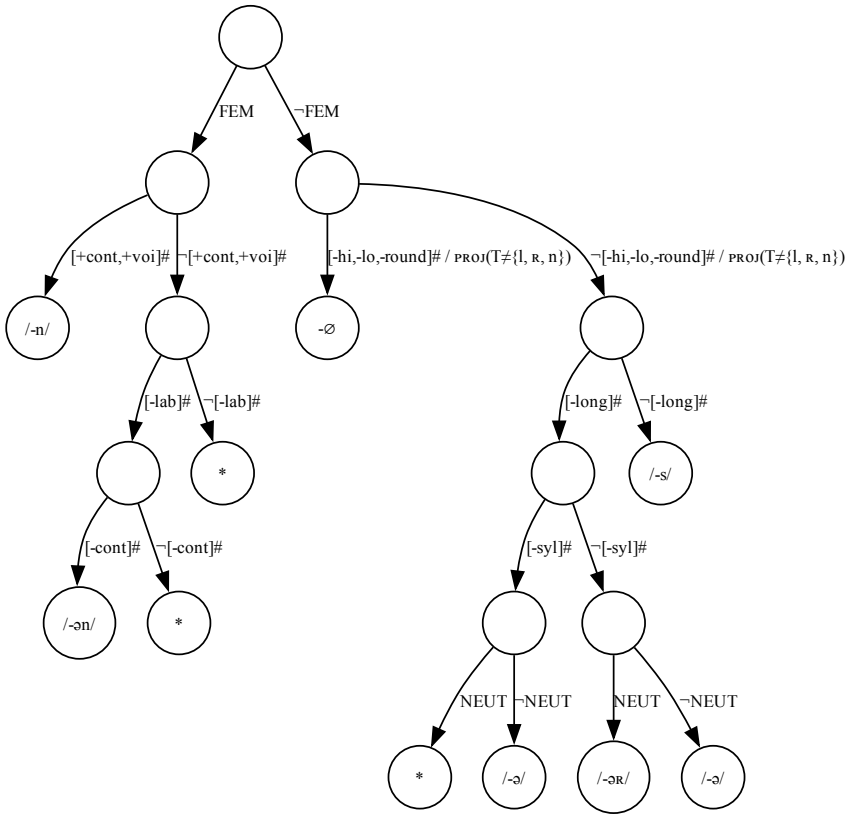


Figure 8: ADP-learned decision tree for German plurals. All five suffixes have productive conditions; "*" marks unproductive conditions. The early split on FEM captures the fact that feminine nouns productively take /-(ə)n/. The condition [-hi,-lo,-round]# / PROJ(T $\neq$ {l, ʀ, n}) is a tier-based generalization of the kind described in Section 3.4: on the tier that excludes {l, ʀ, n}, it picks out words ending in /ən/#, /əʀ/#, or /eːl/#, which take the null suffix (Wiese 1999), and six /ɛl/# words, which do not and are lexicalized as exceptions.

Fig. 7 shows that only ADP produces a distribution of over-applied suffixes analogous to German-learning children. The null suffix, by far the most frequently over-applied suffix by children, is hardly over-applied by any of the other models. Here, the explicit representation of rules learned by ADP is especially helpful for understanding its properties. As in Fig. 8, ADP correctly learns a well-known observation (reduced final syllable constraint; Wiese 1999) that words ending in /ən/, /əʀ/, or /eːl/ typically take null. As discussed in Section 3.4, ADP expands the window of search if the final segment alone fails to yield productive rules. In the present case, just the final segments such as /n/, /l/, and /r/ do not yield any productive condition for the null suffix. Scanning one additional segment, as a last resort, does, effectively discovering three productive rules conditioned on two segments, which applies to the non-suffixed words such as *Teller* ('plate') and *Mädchen* ('girl') in the Laaha et al. (2006) study. Productive rules for other suffixes, at least based on the Leo Corpus (Behrens 2006), are also identified. Yet there are still branches for which no produc-

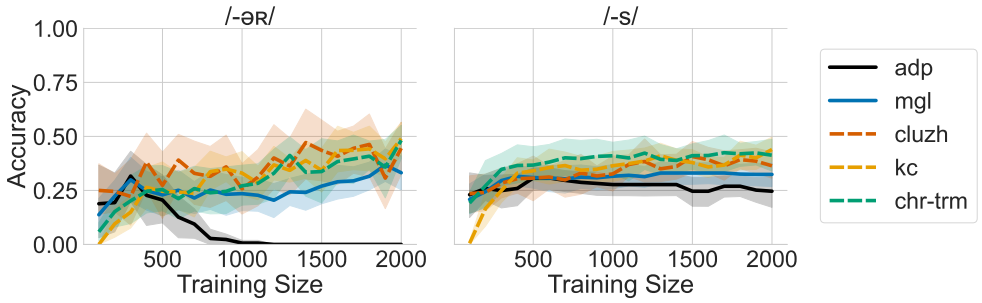


Figure 9: Generalization of the German *-er* and *-s* suffixes as a function of input size.

tive rule can be established (indicated by *): words falling under these branches would surface as unmarked. No model showed significant over-application of the *-s* suffix on the test items in Laaha et al.'s study but all models did in fact learn its productivity. The final grammars across seeds, while extensionally largely uniform, show considerable variation, which corresponds to the individual differences observed in German-learning children (Gawlitzek-Maiwald 1994, Szagun 2001). Fig. 9 focuses on the two least frequent suffixes (*-er* and *-s*). Notably, ADP eventually learns *-er* to be unproductive, which we believe more accurately reflects the mature German plural system, as many studies regard that suffix to be idiosyncratically associated with nouns (Wunderlich 1999, Trommer 2021) including psycholinguistic evidence for whole-word frequency effects (e.g., Penke and Krause 2002, Sonnenstuhl and Huth 2002).

4.3 Icelandic Gender Assignment

The aim of our Icelandic case study is two-fold. On the one hand, the problem illustrates the generality of our formulation of learning: the child is to identify the mapping between input items and output labels – be they suffixes or genders; we return to more general cases in Section 5. On the other hand, just as morphology has productive and unproductive processes, similar dynamics arise for gender assignment (Corbett 1991), which ranges from transparently rule-based to lexically arbitrary. In all cases the learner's task is the same: identify a generalization if one can, and resort to rote memorization if one must.

Gender assignment in Icelandic is especially interesting in this context: Some nouns follow productive rules of gender assignment while others are lexically specific, so that the learners must negotiate the boundary between regularity and arbitrariness. Like many related languages, noun gender in Icelandic is conditioned on the nominative singular, by far the most frequent inflected form. In a recent study, Björnsdóttir (2021) examines the statistical distribution of nouns and their genders from several corpora. According to the TP, the nominative singular suffix *-r* productively predicts masculine, *-i* neuter, and *-a* feminine. However, the null suffix fails to reach productivity for any gender even though neuter is by far the most probable outcome. For example, the child-directed corpus contains 198 nouns with the null suffix: 134 (68%) are neuter, but the 64 masculine and

feminine exceptions exceed the TP threshold ($\theta_{198} = 37$). This productivity prediction holds across corpora including children's naturalistic productions and a large word frequency list. Using a set of novel nouns, Björnsdóttir (2021) confirms that *-r*, *-i* and *-a* indeed productively predict gender as both children (2;9-6;3) and adults assigned the predicted gender categorically. For the null-suffixed noun, however, there was no systematic pattern in either child or adult responses. Furthermore, there is minimal difference in response between children (of all ages) and adults (Björnsdóttir 2021, p1226): the gender system is learned early and accurately, including the corner of the lexicon where no rules are found.

We trained the learning models based on nouns from Björnsdóttir's 2021 study. There are a total of 494 nouns, each transcribed in IPA and is associated with a gender label.[9] As before, input samples ("seeds") were generated stochastically based on the token frequencies of the nouns inflected in nominative singular. The models are tested on the novel noun stimuli from Björnsdóttir (2021) and their predictions are compared against human responses. Fig. 10 provides a very clear illustration of the differing behaviors across the models. ADP, like human subjects, makes categorical predictions of gender for nouns with *-r*, *-i*, and *-a* nominative singular endings. While MGL makes correct predictions for the productive conditions, it almost always, and incorrectly, predicts neuter for nouns with -ø ending: both patterns are consistent with its winner-take-all design. The behavior of the neural models are again difficult to interpret but are clearly at odds with human behavior.

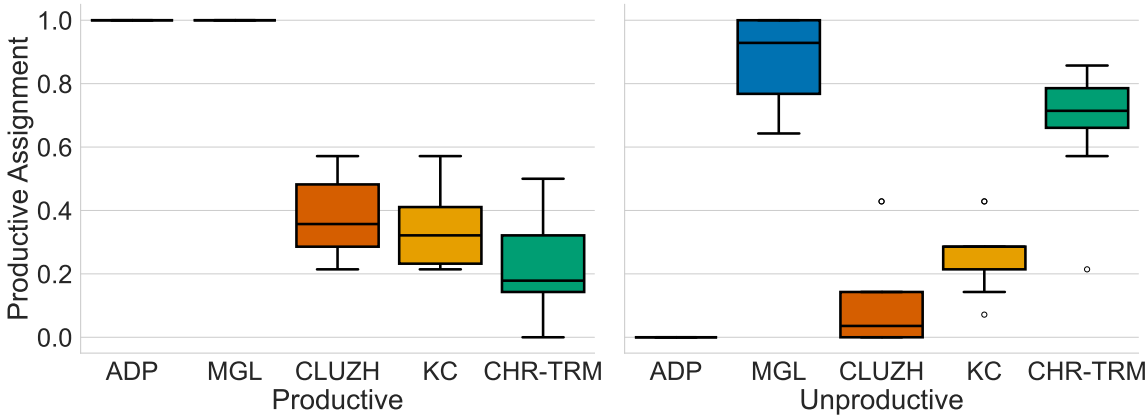


Figure 10: Percent of nonce words in "productive" condition predicted to be majority gender (taking *-r, -i* suffix and predicted masculine or taking *-a* and predicted feminine) and likewise for the "unproductive" condition (taking null suffix and predicted neuter). The results are calculated for models after the full training data; box plots represent variation over the ten seeds.

Before moving on to the two final case studies, an inspection of the terminal state of the learning models provides an illustration of the computational complexity and interpretability of the learning models. As Table 1 shows, ADP learns a handful of immediately interpretable rules as decision

[9]We thank Sigríður Björnsdóttir for providing us with the corpus and for verifying the phonemic transcriptions.

model	English 1,800 words	German 2,000 words	Icelandic 494 words
ADP	4.0 (±0.0)	10.9 (±1.0)	6.5 (±0.5)
MGL	2368.4 (±59.6)	5674 (±92.3)	2479 (±293.1)

Table 1: Complexity of the models of ADP and MGL as measured by the number of rules over the entire datasets.

trees for all three languages. MGL produces many times the number of rules than words: while its design follows the optimizing principle (i.e., selection of the rule with the highest confidence score), its behavior is difficult to interpret for specific instances as the number of rules under competition is very high. As usual, it is difficult to interpret the representation of acquired knowledge and its complexity in neural networks.

4.4 Mandarin classifiers

We now turn to the case of Mandarin classifiers. In Mandarin, classifiers are generally obligatory before the noun when determiners and numerals are used (Li and Thompson 1981). This formal constraint is learned very early by children (Lee 1996). The choice of classifier for nouns, however, is semantically conditioned and highly complex. There are over 100 classifiers: there is a default classifier *gè*/个 but many classifiers are highly specific (e.g., *lún*/轮 for moon and sun) and some have complex conditions that are culturally specific (e.g., *suō*/梭 for a cartridge clip of bullets or *jì*/剂 for a dose of Chinese herbal medicine); see Gao and Malt (2009). Acquisition requires extensive experience: 7-8 year old children manage to approach adult level but only for some of the most common classifiers (Chien et al. 2003).

The semantic basis of classifiers contrasts with previous case studies where the criterion of productivity is phonological (English past tense), grammatical (gender in German plurals), or both (phonological ending of case marker in Icelandic gender). Past acquisition research has asserted the priority of phonological over semantic features in noun class learning (Gagliardi and Lidz 2014, Culbertson et al. 2017). Indeed, recent work on Bantu languages has used the TP to show that, contrary to the descriptive literature, almost no semantic properties yield productive noun classes (Kanampiu et al. 2025, Malone et al. 2026). The phonological forms of words may be easier to learn than meanings, which need to be extracted from highly ambiguous context (Gleitman and Trueswell 2020), but that would be an independent fact about language acquisition, not a substantive bias for sound over meaning for grammar learning. Even if such a bias is true, the work leaves open how and when their roles are reversed — as must be the case for the Mandarin classifiers.

Ancillary assumptions such as substantive biases are not necessary under the ADP model. As illustrated in the coloring example, ADP is a formal algorithm concerned with only the coverage of hypotheses independently of the substantive nature of the hypotheses themselves. Rules are

established, and the corresponding features are invoked, if and only if they provide a sufficiently good coverage of the input data. The phonological features of Mandarin nouns are, of course, available to the child learner alongside the semantic features. Because the ADP algorithm provides a mechanistic process for forming productive rules, any phonologically based generalizations will (presumably) result in failure, while semantic features lead to success.

To illustrate the adaptive nature of the ADP model, we extracted a corpus of the most frequent 111 nouns from the child-directed Mandarin corpora in CHILDES (MacWhinney 2000) along with seven classifiers that are also highly frequent and appear early in child speech (Hu 1993, Erbaugh 2006). These classifiers all have productive semantic conditions, and have previously been used to test young children's classifier knowledge with novel items meeting the conditions (Li et al. 2010). The classifier *gè*/个 covers a majority of 65 nouns yet it falls far short of the TP threshold (87 needed for $N = 110$): again, recursive search for more specific conditions is necessary. The other classifiers are *zhī*/只 (typically for animals such as tiger and body parts such as hand), *tiáo*/条 (for long and slender objects such as snake and river), *zhāng*/张 (for flat objects such as paper and table), *liàng*/辆 (for vehicles such as car and bicycle). Most of these classifiers have exceptions, such as the highly specific classifier for horses (*pī*/匹), which also appears in the list. As in all case studies, the nouns are represented as a sequence of phonemes, with a separate field (with value of 1-4) that marks the lexical tone of the head noun. In addition, the nouns are also annotated with eight binary semantic features: [±human], [±animate], [±body part], [±natural kind], [±concrete object], [±slender], [±flat], and [±vehicle]. Some of these are grounded in the conceptual and perceptual system of pre-linguistic infants (see Carey 2009 for review) while others, such as the concept of a vehicle, cannot be innate but must be accessible to children of appropriate age to be learnable as a semantic condition. For this study, we only present results from the ADP model: the MGL model cannot handle complex non-phonological features as only one additional "segment" can be added (Albright 2002, p72), and the neural networks cannot be effectively trained with such a small amount of data.

As shown in the decision tree in Fig. 11, the ADP correctly identifies the semantic criterion for five classifiers: four have productive conditions with varying level of tolerable exceptions and the other, *gè*/个, is the default classifier which applies when all others fail, with a tolerable number of exceptions of its own. In realistic setting of language acquisition, some of the productive rules are likely learned later. More importantly, no phonological properties are discovered, and not all available semantic features (e.g., concreteness, natural kind) are implicated either: not because they are reflection of learning biases but because they do not lead to productivity in noun classification, a purely quantitative matter. Any attempt to establish productivity along these lines will lead to failure, ultimately guiding the model to the correct features.

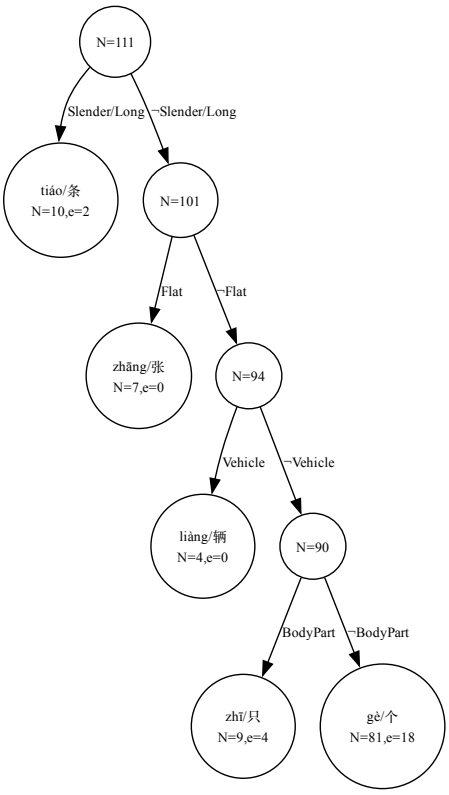


Figure 11: Tree for Mandarin noun classifiers: only semantic features are identified as productive and phonological features are ignored.

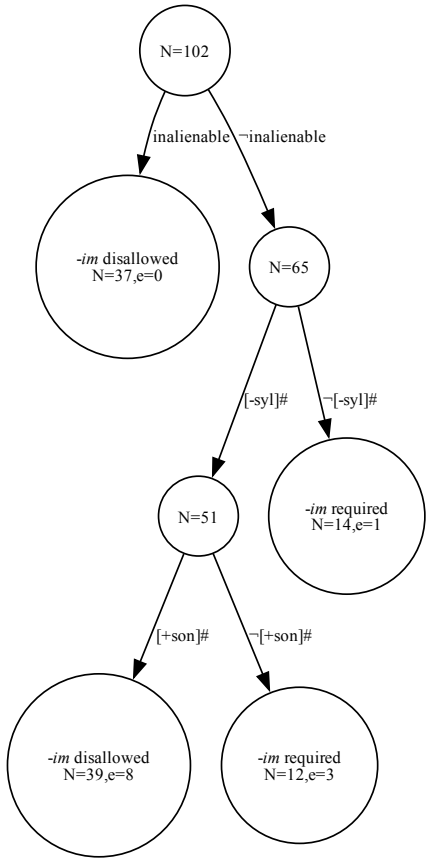


Figure 12: Tree for Northern East Cree. Rules are written to match the descriptions in Henke (2022).

4.5 Northern East Cree

Our last experiment examines the acquisition of the possessive suffix in Northern East (NE) Cree. NE Cree is a variety in the Cree-Innu-Naskapi (CIN) dialect complex of the Algonquian language family (MacKenzie 1982). Of interest is the distribution and acquisition of the *-im* possessive suffix. The suffix is either required or disallowed on the noun, but the decision is based on a complex interaction of factors (Junker et al. 2012, Collette 2014). As discussed below, ADP offers an account of how children acquire the suffix but also provides guidance on the theoretical treatment of the problem.

Nouns in NE Cree are either semantically animate or inanimate. The classification is often based on biological properties but can also be lexically arbitrary. Nouns are also either semantically inalienable or alienable: the inalienable nouns are those of body parts, kinship terms, and items of close personal ownership as is common across languages. Both inalienable and alienable nouns take

a person clitic that marks the possessor. They are distinguished when no possessor is specified: only inalienable nouns must still be marked with an indefinite clitic but alienable nouns are unmarked.

The key problem is to characterize the conditions under which the *-im* suffix is necessary. Due to the complex ways in which (morpho)semantic and phonological properties of the nouns interact, previous research has yielded divergent descriptions; see Henke (2022) for a summary. Junker et al. (2012) state that inanimate nouns generally disallow *-im* while animate nouns generally require it. By contrast, the very detailed analysis of Collette (2014) notes that inalienable nouns disallow the suffix, along with nouns ending in certain segments. Furthermore, English loanwords are categorically alienable regardless of their semantic properties, and they typically require *-im*. And the existence of lexical exceptions has been noted by all authors. The *-im* suffix therefore provides an interesting challenge for models such as ADP, which in turn offers a quantitative and learning-theoretic adjudication of alternative accounts of the language.

The study by Henke (2022) examined 102 nouns from a child-directed corpus of NE Cree (Brittain et al. 2007) and applied the TP to analyze the distribution of the *-im* suffix.[10] Of the nouns, 30 take *-im*: According to the TP, neither presence nor absence is a productive pattern so recursive search in subdivisions is necessary. When approximating the growing lexicon by including the noun types in decreasing order of token frequency, Henke finds that inalienable nouns indeed do not take *-im*. For alienable nouns, those ending in nasals [m, n] do not attach *-im* and those ending in non-nasals do, with a tolerable number of exceptions for both conditions. Henke further provides the longitudinal development of two children learning NE Cree to be consistent with the predictions of the TP. For example, the recursive search procedure predicts that the phonological condition for the presence of *-im* cannot be learned before the morpho-semantic consideration has already taken place, for otherwise there would be too many exceptions (non-*im* nouns) for the detection of productivity. Also, the productivity of *-im* suffixation based on phonology should lead children to over-regularize the exceptions. Both predictions are confirmed.

Our implementation uses the same 102 nouns from Henke's corpus study. As before, the nouns are annotated with their IPA transcriptions along with two additional binary features, [±animate] and [±alienable].[11] The output tree is shown in Fig. 12.

It is clear that the decision tree corresponds to Henke's analysis.[12] Note that the feature animacy is not invoked, in agreement with Collette (2014)'s analysis. Junker et al. (2012)'s description based

[10] Henke also successfully used the TP to identify the distribution of the indefinite possessive clitic and found it to be an effective criterion for distinguishing inalienable and alienable nouns.

[11] We thank Ryan Henke for providing the data and validating our transcription.

[12] There is one alienable noun (*utâpânâskw* 'car') that ends in /kw/; the final segment [w], a sonorant, led to the generalization [+sonorant], whereas Henke postulates a [+nasal] condition, presumably treating this word as an exception. We suspect that Henke's analysis is correct: the [+sonorant] condition learned by ADP holds for the 102 nouns studied here but may be falsified on a larger vocabulary is included if enough [+sonorant] nouns in fact take *-im* and act as exceptions. The ADP model would be able to revise the grammar to the [+nasal] condition. Needless to say, this is an empirical matter that can only be settled when a larger word list becomes available.

on animacy does have some numerical merit: the proportion of *-im*-suffixed nouns is much higher for animate nouns than inanimate nouns (two-tailed Fisher exact test, $p = 0.028$), yet this apparent fact is *undiscoverable* by the learning model. Because the model learns the overwhelming regularity that inalienable nouns resists *-im* very rapidly, the statistical regularity concerning animacy loses out to the still stronger phonological conditions. The decision tree also explains why English loanwords, which are always alienable even if the native counterparts are inalienable (e.g., *mommy*), will generally take *-im*; animacy does not seem to play a noticeable role (Collette 2014, 302). This is so even though the total number of *-im*-suffixed nouns is the minority: subdivision and recursive search enable the productivity of rules under more restrictive conditions as shown in previous case studies.

5 Conclusion

The discovery procedure was already abandoned when introduced to a wider audience (Chomsky 1957) and had completely receded into the background by the time explanatory adequacy became the central goal of the generative enterprise (Chomsky 1965). The path forward from that point on was to define as precisely as possible the substance of Universal Grammar (UG) informed by typological research (Chomsky 1981, Prince and Smolensky 2004). The child's task was no longer discovering a grammar but selecting one out of a finite (albeit large) range of options, a problem naturally suited for the optimization based approach that has been dominant in machine learning and cognitive science.

But the psychological status of optimization has always been questionable. In his pioneering work that led to a Nobel Prize in economics, Simon (1957) observes that individual and organizational decision making often do not strive for optimal solutions because they may not have the information or computational resources required to do so: a satisfactory decision will have to do. This view of *bounded rationality*, to be contrasted with the neoclassical economic assumption that agents are perfectly rational and relentlessly maximize utilities, is supported by an abundance of empirical evidence across social, behavioral and cognitive sciences. But it has not yet supplanted the rationality approach. Critically, while bounded rationality often provides a good description for human behavior, what counts as "good enough" has not been formally specified in a theoretical framework (Russell and Norvig 2004, p. 1049), which has severely limited its utility and predictive power in both formal and empirical research.

The TP provides a precise bound of rationality. Although rarely viewed as such, language acquisition is a paradigm example of decision making under ecological conditions befitting the bounded rationality approach. Every child can learn any language of the world, from only a partial sample of its patterns and, so far as we know, with very limited computational resources. The tension between systematicity and arbitrariness, often resulting from contingencies in language change,

places a limit on the predictive scope of a grammar (Section 1), and the developmental evidence in Section 2 points the same way: children's grammar does not grow monotonically, and conservativity before productivity is what a threshold account of learning predicts, most saliently in gaps where no output form is possible. The ADP model builds the TP into an abductive learning model that iteratively generates, evaluates, and revises hypotheses, incorporating general constraints from the study of learning and reasoning in other domains (Section 3). The case studies in Section 4 confirm its predictions where optimization based models, at least those tested under current configurations, fall short: the distribution of erroneous forms, the asymmetry between over-regularization and over-irregularization, and the absence of an output form when no productive rule can be found. The decision trees produced by ADP map directly onto structural and experimental analyses of language, and its applicability to datasets as small as the 102 nouns of Northern East Cree (Section 4.5) may prove beneficial for the description, preservation, and pedagogy of under-resourced languages. Finally, virtual learners across seeds converge to largely uniform grammars but finer differences may remain for highly complex systems such as the German plurals, which can be regarded as the ultimate source of language variation and change.

As a formal model, ADP is concerned only with the quantitative coverage of the grammar under consideration but not the substantive nature of the grammar itself. The grammar itself, however, has been the focus of linguistic theory after the abandonment of a discovery procedure in favor of an evaluation procedure. The very nature of TP draws a sharp distinction between what is descriptively available in the data (E-language) and what is psychologically available in the grammar (I-language). For example, a majority pattern, a fact of the data, may become noticeable by a linguist and will surely pass a statistical significance test. But as seen in many previous and current studies, unless that pattern also passes the TP threshold, it is not discoverable by the human learner and, by hypothesis, will not become part of the linguistic competence of the speaker. The present study already highlights several points of rupture between the learning-theoretic and structural accounts of language: Constraints of pattern learning have taken the place of locality principles postulated for linguistic representations (Section 3.4), phonological naturalness invoked in theoretical and typological research have developmentally questionable consequences in the acquisition of English past tense allomorphy (Section 4.1.4), and a substantive bias prioritizing phonological over semantic features is not only unnecessary but also has little to say when it must be overridden (Section 4.4). Ideally, the grammar resulting from a discovery procedure that accurately accounts for children's acquisition of language should be identical to the grammar postulated by the linguist based on the terminal state of the speaker's linguistic competence. That is, an explanatorily adequate grammar turns out to be a descriptively adequate one. When the two come into conflict, it is our opinion that explanatory adequacy takes precedence over descriptive adequacy.

It must be emphasized that ADP is only a beginning, and a key component, of a discovery pro-

cedure. It does not operate directly from a corpus of speech but from a list of words and those associated properties. The problem of word segmentation, phonemic categorization, and lexical acquisition must be resolved before the deployment of the ADP model. Nevertheless, the return of a discovery procedure raises questions about the role of UG in language acquisition, and thus its contribution to the goal of explanatory adequacy. Adopting the theory of universals in Chomsky (1965), we propose that formal linguistic universals (hierarchical Merge) are very likely to be innate and domain specific (Smith and Tsimpli 1995, Moro 2008, Berwick and Chomsky 2016). By contrast, substantive universals (e.g., categories of words, the content of rules) are not innate but are constructed in conjunction with other domains of cognition and perception. Even in the morpho-phonological learning problems studied here, the learner must know to search for rules that compose with input to form output – a formal universal – but the representation of rules and their discovery are domain general (recall the coloring example in categorization formation, and indeed its application to phonological rules defined over linear sequences of segments). As succinctly put by Lila Gleitman in the epigraph of this paper, the range of possible grammars is ultimately bounded by the human mind. In our view, no complete inventory of linguistic representations can be fully specified *a priori*. While some, such as animacy, numerosity, and conceptual-intentional content, are likely grounded in the prelinguistic cognitive system (Spelke 2022), others, such as the semantic categories for Mandarin classifiers, can only emerge through social and cultural development. As long as the representational primitives are recursively enumerable, ADP may serve as a regulator and select only those that lead to productive grammatical systems. The learned representations can be used as an input feature to construct additional representations. For example, the category of "noun" in German can be viewed as phonological words which, in conjunction of a preceding marker (i.e., gender-marked determiner; Melançon and Shi 2015), predicts another marker (i.e., the plural suffix). And once the noun plural form is learned, it can be used to predict the form of the verb and the hierarchical configuration that relates them: that is, the language has subject but not object agreement. Such a distributional approach to language, we believe, is the in spirit of the discovery procedure originally conceived (Chomsky 1955). It is the search for productivity, quantified on a purely formal and numerical basis, that determines the ingredients of a language and their assembly.

The pursuit of a discovery procedure naturally raises questions about the nature of language and its unique place in cognition. To put it simply: what can be learned from the data needn't be built as domain specific knowledge and constraints, a guiding theme in the Minimalist Program that aims to reduce the role of genetic endowment for language (Berwick and Chomsky 2016). Ironically, LLMs are in a sense discovery procedures: they are mechanical procedures operating over a corpus and are capable of producing behaviors that approximate the properties of a grammar. Of course, for reasons both known and unknown, LLMs are unlikely to be the kind of discovery procedure that children use to learn and linguists seek to understand. The impetus, now more urgent than ever, is

to develop a discovery procedure of the right kind, at least one that is good enough.

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